Risk Premia in Diversified Energy Portfolios

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Abstract

We study the behavior of classical risk premia in energy futures, focusing on momentum, carry, value, and statistical arbitrage across a broad and updated commodity universe. We show that while some premia have weakened in traditional benchmark contracts, others persist or re-emerge in newly liquid markets and in signals that had been largely inactive in prior periods. At the portfolio level, low correlations across economically distinct strategies generate strong risk-adjusted performance under simple aggregation. We then examine the practical mechanics that govern real-world implementation, including roll dynamics, term-structure effects, and portfolio integration. Together, the results provide an implementation-aware perspective on how energy risk premia behave today and how they can be combined into robust systematic portfolios.

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1. Introduction

Risk premia, the return earned for taking on structural risk in a market, have long been a source of systematic profit across asset classes. In energy futures, these premia reflect persistent features of physical markets, the behavior of market participants, and expectations about future supply and demand. These observable forces can be quantified with data and turned into systematic trading signals, a practice studied by quantitative researchers at trading houses, hedge funds, and banks for decades.

We revisit these structural risk premia in the context of today's energy markets, focusing on how their behavior has changed relative to earlier decades and prior empirical studies. We examine what the evolution of these signals reveals about market structure and consider the implications for their practical use. Some signals have become more widely traded and more heavily modeled, raising the possibility of alpha decay and trade crowding. At the same time, new commodities and previously overlooked fundamental relationships may still carry premia that have not been fully exploited.

We study classical momentum, carry, and value strategies across a broader set of energy commodities and evaluate how these premia interact at the portfolio level. We emphasize contract selection through rolling methodology, the influence of market structure, and the underlying physical relationships each signal is intended to capture. Our objective is to assess whether widening the universe of commodities and signals leads to a more balanced and reliable energy risk premia portfolio in a changing market environment.

2. Risk Premia in Energy Commodities: Classical Signals

Energy markets produce several distinct forms of risk premia due to real operational constraints, including production levels, transportation limits, storage availability, and the hedging needs of physical market participants. These economic forces are reflected in futures prices and in the shape of the contract term structure. As a result, certain patterns in returns appear repeatedly, even as market conditions evolve. These patterns vary in their economic origin and in the part of the curve from which they draw information, which is why they respond differently to liquidity, roll choice, and regime shifts. We focus on the three classical premia that form the foundation of most systematic energy strategies: momentum, carry, and value.

2.1 Momentum

Time-series momentum, in its basic form, is a trading strategy that follows the direction of the price trend. A momentum strategy takes long positions when recent prices are rising and short positions when they are falling. Momentum is one of the simplest, and therefore one of the most widely used systematic signals.

In energy markets, momentum reflects slow-moving physical supply and demand adjustments. For example, in the short run, oil demand is price inelastic, as buyers continue to purchase at higher prices because the underlying need for the product is not easily replaced. On the supply side, production, refining activity, transportation, and storage all adjust with meaningful delays, as physical operations require time and capital to scale up or down and cannot be paused or restarted instantaneously. When inventories draw or build, these physical imbalances tend to persist, and prices adjust gradually rather than instantaneously. A trader who aligns with this adjustment process is effectively compensated for bearing the risk that physical imbalances persist, with sustained directional price movements providing that compensation.

Momentum, in this context, is not an anomaly or a purely statistical pattern. It arises from structural features of the market, where physical constraints cause information about supply and demand imbalances to be incorporated into prices only gradually, generating persistent directional moves that trend-following strategies can capture.

It is important to highlight a distinction regarding the definition of this signal. Momentum appears in two forms throughout commodity quantitative research literature:

- Time-series momentum, which evaluates each asset independently and takes long or short positions based on its own recent return, and
- Cross-sectional momentum, which ranks assets relative to one another, buying the most upward-trending assets and selling the most downward-trending assets.

Cross-sectional momentum is effectively a ranking mechanism: it identifies the commodities with the strongest recent trends relative to the rest of the basket. Our goal, however, is to combine signals across commodities within a portfolio, where relative performance will naturally emerge through position sizing and correlation structure. Studying time-series momentum directly is therefore more informative. It captures the trend in each market on its own terms rather than filtering trends through cross-sectional comparisons, hence we will refer to momentum as the time-series version throughout this research.

In its most basic form, the momentum signal can be applied directly to the price of futures contracts. Let F_t denote front month settlement price at time t.

We introduce a price averaging component MA_t, defined as:

$$MA_t(F;n) = \frac{1}{n}\sum_{i=1}^{n} F_{t-i+1} \ ,$$

and the long/short signal for momentum effectively becomes:

$$\pi_t^M(F) = \begin{cases} +1, & if F_t - MA_t(F;n) \geq 0 \\ -1, & if F_t - MA_t(F;n) < 0 \end{cases}.$$

There is an endless list of other price and price-like features upon which one can apply a moving average to produce a similar trend-following signal. While we avoid diving too deeply into these variations so as not to overfit our estimates at this stage, it is important to make two minor adjustments to the basic signal to obtain more stable results.

The first adjustment is to apply the signal not directly to raw price data, but to a roll-adjusted futures time series. We will explain the changes that come with rolling futures in a later section, but the intuition is straightforward: roll-adjustment helps weed out false momentum that arises purely from seasonal price patterns rather than genuine trend.

As an intuitive example, consider the RBOB gasoline futures contract. The underlying product exhibits strong seasonal behavior, with prices rising each spring and summer as driving demand increases. A simple price-based momentum rule would therefore generate long signals every March–April, not because of genuine trend, but because of this predictable seasonal pattern. Without roll-adjustment, such seasonality would systematically contaminate our trend estimate.

The second adjustment is to average the momentum signal across several values of the lookback parameter n. One way to introduce this averaging directly into the signal construction is to use a two-moving-average crossover, which serves as a composite indicator of trend strength. For two horizons m and n, we define

$$M_t(m,n) = MA_t(F;m) - MA_t(F;n), \qquad with\ m < n$$

The intuition behind this crossover signal is to incorporate a more holistic view of price dynamics. A long position corresponds to cases where prices are, on average, rising relative to their longer-term history, while a short position reflects prices declining relative to that history. By comparing a short-horizon moving average to a longer-horizon one, the signal captures whether recent behavior is meaningfully stronger or weaker than the broader trend. This helps filter out noise: isolated short-term movements that are inconsistent with the longer-term pattern will not immediately flip the signal, whereas sustained deviations in either direction will.

Bringing this together, the composite momentum signal is constructed as the sign of the sum of three short-term crossover signals. Let (m_1, n_1), (m_2, n_2), (m_3, n_3) denote the three chosen parameter pairs. Our aggregated momentum signal is then defined as:

$$MA_t^{agg} = \left(\frac{1}{3}\right)[\pi_t^M(F;m_1,n_1) + \pi_t^M(F;m_2,n_2) + \pi_t^M(F;m_3,n_3)]$$

$$\pi_t^{M,agg}(F) = \begin{cases} +1, & \text{if } MA_t^{agg} \geq 0; \\ -1, & \text{if } MA_t^{agg} < 0 \end{cases}$$

Each sub-signal takes values in $\{+1, -1\}$, and applying the sign of their average ensures that the aggregated momentum signal also remains within this set. This construction treats the combined signal as a single momentum strategy and keeps the resulting position fully long or fully short at the strategy level. While fractional sizing across signals is a natural alternative and is introduced later at the portfolio construction stage, we retain a binary specification here to define a single composite momentum signal. The specific parameter choices (m_j, n_k) are specified later in the Data section.

We average our momentum signal because the momentum lookback parameter n is notoriously sensitive to overfitting. For reference, we present momentum performance across a range of n values in Appendix A. By combining crossover signals at multiple frequencies, we capture a broader and more stable representation of underlying trends, moving closer to the true essence of the signal and further from the risk of tailoring our results to in-sample strength.

Several additional characteristics of momentum are important in practice. While the signal performs well over long horizons, it also exhibits extended periods of underperformance. In the Results section, we show that such drawdowns recur over time and have become more pronounced in recent years. One plausible explanation is trade crowding, as momentum strategies grew significantly in popularity, raising the possibility of alpha decay. Momentum also struggles during prolonged range-bound, directionless market environments in which no clear trend emerges. Intuitively, such conditions are challenging for trend-following strategies, as they repeatedly enter and exit positions without capturing sustained price moves.

These observations motivate the refined, "second-generation" systematic risk-premia signals introduced later, which are designed specifically to mitigate these weaknesses.

2.2 Carry

The most intuitive way to think about systematic carry is that it earns the return associated with holding a position when the market itself does not change. In commodities, this return arises from roll yield; the gain or loss generated by holding and rolling futures along the term structure. The classical theory of storage decomposes roll yield as

$$j = y - u - r,$$

where y denotes the convenience yield, u, the storage costs, and r, the financing rate. If the convenience yield exceeds storage and financing costs, the curve is backwardated, with deferred futures priced below spot. A long futures position in backwardation then earns positive roll yield, our "carry," as deferred prices converge upward toward spot. When storage costs dominate, the curve is in contango and deferred futures trade above spot. Contango carries a natural negative roll yield as prices converge downward, which motivates taking the short side of futures. A carry strategy seeks to systematically harvest this convergence return implied by the curve's existing shape.

A complementary way to understand the carry risk premia is through inventory flows. The dominant commercial actors in crude oil markets are inventory hedgers, primarily producers and large-scale physical holders, who are naturally short barrels because their business model involves selling physical crude. These hedgers routinely hedge this exposure by locking in forward prices, even when those prices are unfavorable relative to spot. In doing so, they transfer the benefits or costs implied by the shape of the curve to whoever takes the opposite side. A financial trader who goes long in backwardation effectively receives the premium that a producer gives up to secure price certainty, while a trader who goes short in contango assumes the role of absorbing storage-related price risk that commercial firms prefer to avoid. In this sense, carry risk premia reflects the return earned for providing a form of insurance for these inventory hedgers.

This economic intuition motivates our empirical implementation of carry, which we define using the slope of the futures curve to take long positions in backwardated markets and short positions in contango.

$$\pi_t^C(M,N) = \begin{cases} +1, & if\, F_{t,M} - F_{t,N} > \varepsilon; \\ -1, & if\, F_{t,M} - F_{t,N} < -\varepsilon; \\ 0, & if \left|F_{t,M} - F_{t,N}\right| \le \varepsilon \end{cases} \qquad with\ M < N$$

Here, ε serves as a threshold that filters out economically insignificant slopes. Intuitively, steeper carry signals are more reliable because they imply a larger roll yield accumulated between deferred and prompt futures. The threshold therefore avoids trading carry signals that are too flat to reflect meaningful curve structure. While the choice of ε can mechanically affect position frequency and volatility, carry signals are relatively stable across a wide range of reasonable thresholds. As a result, the threshold parameter primarily governs risk exposure rather than acting as a finely tuned driver of empirical performance.

A practical nuance in constructing carry is that it is most reliable when measured over annual horizons, such as comparing F_1 to F_{13}, to avoid the seasonality inherent in many commodity curves. For example, as noted earlier with RBOB, an accurate assessment of the curve's slope requires comparing March across years rather than adjacent contracts.

Despite its economic grounding, carry has meaningful limitations. Distorted, "smile"-shaped curves, which arise episodically in energy markets, may generate signals that do not align with carry's economic intuition across all horizons. Moreover, as we show in the Results section, markets with weak or sporadic hedging pressure, such as natural gas liquids (NGLs), exhibit substantially weaker roll-yield compensation.

2.3 Value

The traditional value risk premia is inherently a trade on mean reversion toward a long-run equilibrium. The idea is straightforward: assess whether the current price of a commodity is above or below its implied equilibrium level and take positions that expect prices to revert toward that level. A long position is taken when the asset appears undervalued, and a short position when it appears overvalued. This intuition is rooted in the classical notion that prices tend to gravitate toward levels consistent with supply and demand balance.

There is no single, fixed definition of value in energy markets, as the level toward which prices revert depends on evolving market structure. As a result, prior work in the academic and practitioner literature reflects a range of approaches rather than a unified framework. Changes in trade flows, benchmarks, regulatory regimes, and supply dynamics can all reshape what constitutes fair value over time, making value inherently more context-dependent than short-horizon price-based signals.

One common perspective defines value as mean reversion toward a long-run real equilibrium, with prices viewed as expensive or cheap relative to their inflation-adjusted historical average. Another popular approach interprets value through long-horizon return reversal, where commodities that have appreciated strongly over several years are considered overvalued and those that have declined are considered undervalued. These definitions of value emphasize slow reversion toward fundamental anchors and can resemble long-horizon analogues of momentum-based signals.

Importantly, the location of this anchor is not fixed and can shift as market structure evolves. A clear example of a shifting value anchor is WTI crude following the lifting of the United States crude export ban in December 2015. The regulatory change altered the role of WTI within the global market, transforming it into the primary benchmark for U.S. production and the Cushing pipeline network rather than a largely landlocked domestic reference. As a result, the level toward which WTI prices reverts to reflects a different set of constraints than in earlier periods, when U.S. crude was effectively isolated from global waterborne markets and Brent served as the dominant international benchmark. This illustrates a central feature of value in energy futures: long-run levels evolve with market structure, and practical value signals must allow the anchor itself to adjust over time.

Given these challenges, we synthesize these multiple perspectives on value to build an indicator that remains robust across time. We define value as:

$$\pi_t^V(F) = \begin{cases} +1, & if F_t - MA_t(F;n) < -\varepsilon; \\ -1, & if F_t - MA_t(F;n) > \varepsilon; \\ 0, & if |F_t - MA_t(F;n)| \le \varepsilon \end{cases}.$$

We impose two additional design choices to ensure that the value signal captures only economically meaningful dislocations. First, we rely on long-horizon moving averages so that value is defined relative to a slowly evolving reference level rather than short-run price fluctuations. Second, we introduce a wide ε threshold band, restricting positions to periods in which deviations are sufficiently large to plausibly reflect genuine mispricing. Unlike carry, where thresholding primarily refines signal quality, the ε band is a central element of the value thesis and is intended to isolate rare, high-conviction dislocations. Consistent with this objective, value positions are implemented further out on the term structure of the futures curve, where carry effects are less punitive and price signals better reflect longer-term fundamentals.

When implemented in this way, value acts as a slow-moving complement to momentum and carry. Rather than attempting to identify short-term turning points, the signal activates only after large and persistent deviations have emerged, capturing gradual normalization as shocks dissipate and market conditions evolve toward more typical states. By responding only to substantial dislocations, the value signal avoids premature mean-reversion bets and contributes stability to a diversified strategy mix.

The primary limitation of value in futures markets arises in regimes with steep carry, defined by a large absolute roll yield, whether in steep contango or backwardation. In such environments, the cost of holding futures positions can be large relative to the expected speed of price normalization. Prices may appear rich or cheap relative to long-run benchmarks, yet the accumulation of carry over the holding period can dominate eventual mean reversion. As a result, value trades become highly sensitive to timing, requiring that normalization occur sufficiently quickly to offset adverse carry, particularly at shorter maturities where carry effects are most pronounced.

For these reasons, many practitioners avoid short-horizon implementations of value altogether and instead extend both the horizon and selectivity of their signals. Properly constructed, value therefore functions as a long-horizon dislocation signal that engages only when prices deviate materially from plausible fundamental anchors, rather than as a frequently trading mean-reversion rule.

3. Risk Premia in Energy Commodities: 2nd Generation Signals

We now introduce a newer class of systematic signals, which we refer to as the "second generation" of risk premia. The aim of these strategies is to move beyond what has traditionally worked in established energy markets and to identify persistent returns in both new commodities and refined signal combinations. As we will show, momentum applied to newly liquid markets such as NGLs can generate strong risk-adjusted returns. It also provides diversification to conventional trend-following portfolios, whose exposures have grown increasingly correlated through widespread CTA adoption. Carry has also performed well, but its recent stagnation reflects a closer alignment with its underlying economic foundation rather than the discovery of new assets. Finally, we refine the traditional value risk premia along two dimensions: first, as an enhancement to momentum portfolios, and second, by redefining value in terms of commodity spreads rather than outright prices.

3.1 NGL Momentum

We now turn to natural gas liquids (NGLs) and examine three of the most liquid contracts in this complex: propane, butane, and ethane. Although we evaluate all classical risk premia on these markets, our emphasis is on momentum, which is particularly strong across all three contracts and exhibits distinctive behavior relative to more established energy benchmarks.

Natural gas liquids are hydrocarbons extracted primarily from natural gas processing and, to a lesser extent, from crude oil refining. They span a range of carbon numbers and physical properties: ethane (C2) is predominantly used as a petrochemical feedstock, propane (C3) serves heating and petrochemical demand, and butane (C4) is used in gasoline blending and as a petrochemical input. Other NGLs such as isobutane and natural gasoline (C5+) are commercially important but lack the liquidity and futures depth required for systematic strategies.

Futures liquidity in propane, butane, and ethane begins to reach reliable depth only after the mid-2010s. For this reason, our analysis begins in 2015, the earliest point at which these contracts exhibit consistent volume necessary for robust back testing. This timing also marks the period when NGLs became investable for systematic portfolios rather than niche physical markets. A summary of NGL liquidity over time is provided in Appendix B.

We define the momentum signal for NGLs identically to the classical specification used earlier. This allows a clean comparison between NGL momentum and its counterparts in the major energy benchmarks, and isolates differences in performance to market structure rather than differences in signal construction.

3.2 Short Term Momentum, Long Term Value

Short-horizon momentum and long horizon value can be viewed as two distinct but complementary risk premia, each capturing a different dimension of predictable dynamics in energy futures markets, whose interaction gives rise to a composite momentum–value effect. Short-term momentum exploits the empirical persistence in nearby contract returns, while long-term value targets slower-moving, larger deviations from equilibrium that have historically exhibited mean reversion.

Deploying these signals jointly allows the investor to harvest short-term continuation while mitigating exposure to persistent overvaluation or undervaluation regimes, giving rise to a conditional risk premium that differs from either component in isolation. In this sense, value serves as a long horizon complement to momentum. Short-run returns in energy markets often display positive autocorrelation as shocks propagate through prices, while longer-horizon price dynamics tend to mean-revert as physical supply and demand adjust.

This complementarity arises from the underlying economics of physical commodity markets. Inventory responses, production adjustments, transportation flows, and refinery optimization operate on multi-month or multi-quarter horizons. As a result, short-term shocks can generate momentum even as the slower supply–demand response eventually restores equilibrium. Momentum and value therefore coexist naturally: one reflects high-frequency price continuation, while the other captures gradual correction toward fundamentals

Operationally, the joint approach behaves like a trend-follower whose position sizing is disciplined by a slow-moving valuation anchor. When a commodity is both trending and undervalued (or overvalued for a short), the valuation component amplifies the momentum exposure by indicating that the directional move is aligned with a return toward fair value. When momentum points in a direction inconsistent with fundamentals, the valuation signal curtails exposure and prevents the strategy from extending into historically unsustainable dislocations.

We therefore formalize a joint momentum–value signal that represents a composite risk premium arising from the interaction of short-horizon trend persistence and long-horizon mean reversion.

$$\pi_t^{MV} = \omega\pi_t^M + (1-\omega)\pi_t^V \ .$$

Here, our ω represents the weight allocated to the momentum leg, with $0 \leq \omega \leq 1$ and $\omega - 1$ allocated to the value leg. Under equal weighting, $\omega = 0.5$ assigns half the exposure to each component. When momentum and value signals point in opposite directions with similar strength, their contributions partially offset, reducing net directional exposure.

Importantly, this offsetting occurs at the signal level rather than constituting a mechanical hedge. Because momentum and value are implemented on different maturities of the futures curve, with momentum expressed in shorter-dated contracts and value in longer-dated tenors, opposing signals do not generally result in a zero-risk or zero-return position. As we show in the empirical results, the blended strategy retains meaningful exposure, reflecting differences in contract dynamics across the term structure.

For this composite signal, contract rolling methodology introduces an additional degree of freedom. The impact of both momentum and value signals depends on the chosen tenors, particularly in markets with pronounced seasonality or where nearby contracts exhibit heightened volatility. These implementation details materially influence how shocks propagate across the curve, and we revisit their empirical consequences in a later section.

3.3 Statistical Arbitrage: Value on a Spread

Statistical arbitrage, often referred to as "pairs trading", is rooted in the idea that certain price spreads exhibit mean-reverting behavior even when the individual futures prices themselves do not. When two commodities are economically linked, temporary dislocations around their fair-value relationship can be viewed as opportunities: one can go long the undervalued leg and short the overvalued leg, profiting as the spread converges back toward equilibrium. In this sense, statistical arbitrage can be interpreted as a value risk-premium applied to a spread rather than to a single asset.

Although statistical arbitrage has a long history, most notably in equity market-neutral strategies developed in the 1980s, our formulation reflects a modern interpretation tailored to commodity markets. Traditional approaches typically rely on purely statistical criteria, such as correlation or cointegration, to identify candidate pairs. In contrast, we adopt a "quantamental" framework in which each spread is motivated by a well-defined economic transformation.

The pairs in this study are selected because they correspond to fundamental commodity arbitrages: storage across time (calendar spreads), movement across locations (location differentials), or processing through refining (crack spreads). Each spread can be interpreted as a real option on a physical asset. For example, a crude–product spread reflects the option to convert crude oil into refined products subject to refinery capacity and operating costs, while a location spread reflects the option to transport barrels across regions when price differentials exceed logistics costs. A full economic justification for each spread is provided in Appendix C. For completeness, Engle-Granger cointegration tests for each pair are reported in Appendix D, though statistical significance is treated as a diagnostic rather than a selection criterion.

The choice to root pairs in economic linkage is critical because commodity futures rarely exhibit a stable, time-invariant equilibrium. Contract specifications differ across markets, infrastructure shifts alter optionality, and structural regime changes frequently redefine what "fair value" looks like. In such environments, the relevant question is not *"Are these two prices statistically related?"* but rather *"Is the current gap between these two economically linked contracts reasonable given the underlying fundamentals?"* The presence of a transformation relationship provides a more reliable foundation for spread construction than stationarity metrics alone.

Formally, for two futures prices $F_t^{(i)}$ and $F_t^{(j)}$, we define the spread:

$$S_t = F_t^{(i)} - F_t^{(j)}$$

And the trading signal as:

$$\pi_t^V(S_t) = \begin{cases} +1, & if\, S_t - MA_t(S_t; n) < -\varepsilon; \\ -1, & if\, S_t - MA_t(S_t; n) > \varepsilon; \\ 0, & if\, |S_t - MA_t(S_t; n)| \leq \varepsilon \end{cases}$$

Here, $MA_t(S_t; n)$ is an n – day backward-looking moving average that serves as our fair-value proxy, and ε is the deviation threshold.

Choosing n, ε, and the rolling schedule introduces more degrees of freedom than any other strategy class. Even for a pair that is economically linked, different contract tenors can be driven by distinct microstructure forces; short-horizon convergence may represent a fundamentally different signal than long-horizon convergence; and deviation bands may be defined in dollars, percentages, or volatility-normalized units. Moreover, spread behavior itself is highly regime-dependent, complicating any attempt to maintain a single time-invariant fair-value model.

A simple example illustrates these challenges. Consider the Brent–WTI spread, which is often treated as a canonical mean-reverting pair. Prior to the lifting of the U.S. crude oil export ban in December 2015, the two contracts were essentially disjoint, with the spread frequently trading near −\$10/bbl, reflecting persistent segmentation between inland U.S. crude markets and seaborne international benchmarks during the shale expansion. Following the removal of export restrictions and the integration of U.S. supply into global trade flows, the same spread has typically fluctuated within a much narrower range, often closer to −\$2/bbl.

A fixed deviation band that works well in one regime therefore becomes either too permissive or too restrictive in another, while the moving average itself may no longer represent a meaningful fair-value benchmark once the underlying market structure has shifted. These dynamics highlight why spread-based strategies require careful interpretation: what may appear to be a tradeable statistical anomaly can, in reality, reflect a persistent transformation in market structure.

The remaining challenge is avoiding excessive discretionary tuning. In principle, a CTA could tailor parameters for each individual pair, given the distinct mean-reversion strengths and convergence mechanics involved. For the purposes of this study, we deliberately take the opposite approach. After identifying economically meaningful spreads, we impose a single, standardized parameter set across all of them. While this choice will not extract the maximum risk-adjusted return from each individual spread, it provides a realistic and disciplined view of how the strategy behaves when aggregated into a single statistical arbitrage portfolio.

4. Data

All data used in this study are daily time series sourced from Bloomberg. We define two subsets of our commodity universe. The first is a group of highly liquid energy futures, referred to as the Core-6: WTI (CL), Brent (CO), RBOB gasoline (XB), Heating Oil (HO), Gasoil (QS), and Natural Gas (NG). The second subset consists of natural gas liquids (NGLs): Propane (BAP), Ethane (CAP), and Butane (DAE), which are treated separately due to their distinct microstructure and more recent transition to deep liquidity.

For each commodity, we source the full strip of generic futures contracts (e.g., CL1 Comdty through CL30 Comdty), providing up to thirty consecutive monthly tenors per asset. This structure allows us to construct consistent futures curves and to express strategies at multiple points along the term structure.

The full dataset begins on January 1, 2010. The formal backtest window for all strategies runs from January 2, 2015 through December 1, 2025. This start date reflects the period in which ICE-listed NGL futures achieved sufficient liquidity for systematic trading following their introduction and growth in 2014–2015. Data from 2010 through 2014 are used exclusively to initialize longer-horizon signals, ensuring that all strategies enter the test window with comparable historical depth.

The raw data are cleaned in two steps. First, we remove duplicate observations that appear on exchange holidays in Bloomberg's generic futures series. Second, we eliminate dates on which CME and ICE calendars do not overlap, retaining only trading days when all contracts in the universe are simultaneously open. This strict calendar alignment prevents artificial suppression of volatility caused by asynchronous holiday gaps and ensures that momentum, roll, and spread estimates reflect genuine market dynamics rather than data artifacts. Any omitted ICE return on these isolated dates is naturally incorporated into the next available trading day's price.

All futures curves are constructed using a standardized end-of-month rolling rule. Contracts are rolled out of the front month once they enter their expiry window, based on empirical expiry calendars obtained from Bloomberg. If the standard end-of-month roll would place the position into a contract already within its expiry window, the roll is executed one month earlier. This procedure synchronizes rolling schedules across commodities and reduces liquidity and delivery risk associated with holding near-expiry contracts.

Some NGL futures are settled against floating or index-based prices, where the front-month contract represents an arithmetic average of daily reference prices over the delivery period. In these cases, the designated front month becomes progressively illiquid as the settlement price is revealed over time. Consistent with market practice, we therefore exclude the front month as a tradable contract for such instruments and instead roll into the nearest actively traded forward tenor.

To place all futures on a common scale, prices are converted to USD per barrel using standard unit transformations, which are detailed in Appendix E. To ensure consistency in return calculations across strategies and to accommodate periods in which futures prices may be negative, we apply a constant shift factor K at the strategy level so that all profit-and-loss series remain strictly positive. The constant is chosen as the minimum value required to ensure positivity across all assets and dates within a given strategy. This transformation is purely mechanical and does not affect relative performance comparisons within a strategy.

All exposures and performance metrics are computed using log returns. Reported "returns" correspond to the mean of log returns, while "volatility" is defined as the standard deviation of log returns. The information ratio (IR) is computed as the ratio of mean log return to log-return volatility, analogous to a Sharpe ratio with the risk-free rate set to zero, as all strategies are implemented using self-financing futures positions.

4.1 Parameter Selection

Strategy	Core Parameters
Momentum	MA crossovers: (1,5), (5,20), (10,60)
Carry	Tenors: F4–F15, ε = 0
Value	n = 10 years (running avg), ε = 10%, M12-13 roll & exposure
Value–Momentum Blend	Weighting: ω = 0.5 (50–50 value/momentum)
Statistical Arbitrage	n = 20 days, ε = 10%

Table 1 Parameter choices used in strategy backtests.

All strategy parameters are fixed ex ante and held constant across assets. Momentum is implemented using a small set of short-horizon moving-average crossovers designed to capture near-term trend persistence. Carry exposure is taken over tenors F4–F15 to ensure consistency across rolling conventions, particularly for NGL contracts whose exposure naturally extends into later months. Value is defined relative to a slow-moving ten-year running average, evaluated using information available up to $(t-1)$; the full ten-year window is reached in 2020 following an initialization period beginning in 2010. For both value and statistical arbitrage strategies, the threshold parameter ε represents a percentage deviation from the relevant reference level, restricting exposure to large, economically meaningful dislocations. Statistical arbitrage employs a short correction window combined with a wide deviation band to focus on pronounced spread dislocations.

5. Results

This section summarizes the performance of the strategies described thus far. We focus on portfolio-level results for each strategy family, with individual contracts or spreads combined using equal weighting unless otherwise noted. This aggregation reflects the practical implementation of these strategies as diversified portfolios rather than as isolated single-asset trades. Detailed commodity-level results are provided in Appendix F.

To assess robustness across market environments, we report performance statistics over both the full sample and subsamples defined by a structural break on January 2, 2022. This date coincides with a pronounced regime shift in energy markets associated with the global pandemic recovery, the Russia–Ukraine war, and resulting changes in inventory dynamics, volatility, and cross-market linkages. Splitting the sample in this manner allows us to examine how strategy behavior and relative performance evolve across materially different economic conditions.

Strategies		Full Sample	2015-2022	2022-2025
CORE-6 MOM	Return	-2%	0%	-4%
	Vol	19%	18%	20%
	IR	-0.08	0.00	-0.22
CORE-6 CAR	Return	7%	8%	5%
	Vol	22%	24%	17%
	IR	0.31	0.31	0.30
NGL MOM	Return	5%	6%	2%
	Vol	5%	5%	4%
	IR	0.99	1.23	0.46
ALL 9 VAL	Return	3%	3%	4%
	Vol	20%	20%	21%
	IR	0.15	0.14	0.17
VAL-MOM	Return	4%	4%	3%
	Vol	8%	8%	10%
	IR	0.47	0.59	0.32
STAT ARB PORT	Return	7%	9%	4%
	Vol	8%	9%	5%
	IR	0.90	0.93	0.92
EW PORT	Return	5%	6%	3%
	Vol	5%	5%	5%
	IR	1.01	1.18	0.65

Table 2 Split-period results across strategies, all strategies equally weighted. STAT ARB PORT corresponds to an equally weighted portfolio of 8 spreads of interest.

5.1 Momentum: Results

Several patterns emerge from the momentum strategies. First, Table 2 shows that traditional momentum applied to the Core-6 energy futures performs poorly over the full sample and deteriorates further in the most recent market environment. This outcome is consistent with the instability of short-horizon trend-following strategies during prolonged periods of range-bound price behavior, as observed after 2022. Momentum depends on the persistence of directional price trends; when markets move sideways for extended periods, frequent trend reversals generate whipsaw losses that materially weaken signal effectiveness.

In contrast, momentum applied to NGL futures remains profitable across subsamples, albeit with reduced performance after 2022. This difference is consistent with stronger structural trends and less crowded participation in NGL markets. These distinctions are illustrated in Figure 1, which plots the cumulative performance of the momentum portfolios over the full sample.

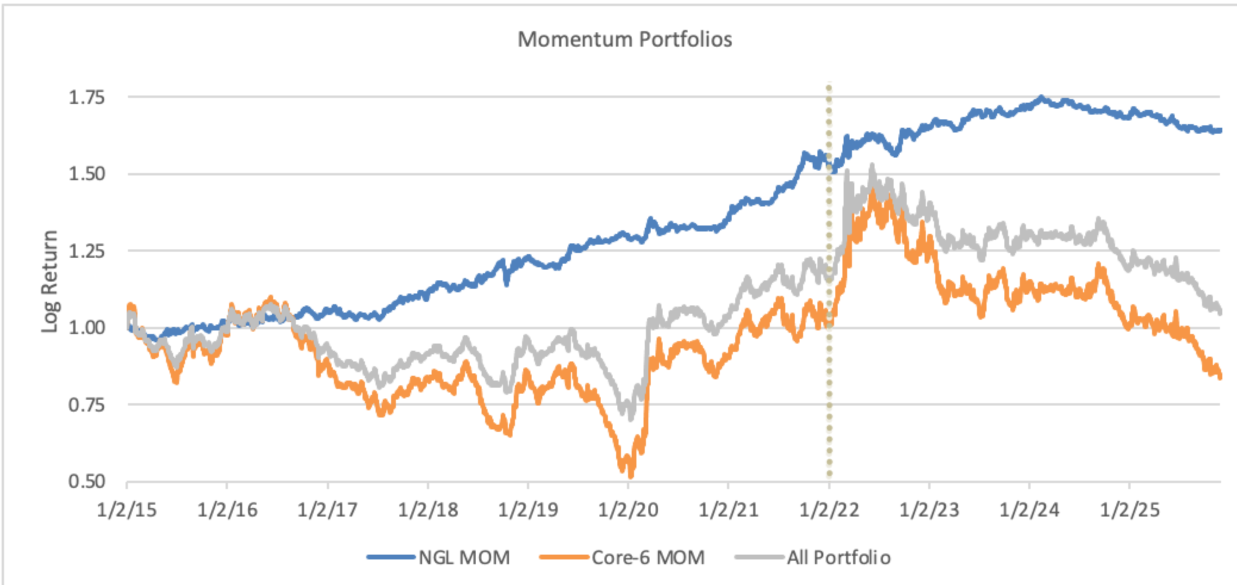

Figure 1 Momentum performance for an equally weighted portfolio of Propane, Ethane, and Butane (NGLs) compared with that of the Core-6 energy futures (WTI, Brent, RBOB, ULSD, Gasoil, and Natural Gas) over the full sample period. The "All Portfolio" series represents an equally weighted momentum portfolio across all nine commodities. The vertical dotted line marks January 1, 2022.

Momentum applied to the Core-6 exhibits pronounced drawdowns and extended periods of stagnation, particularly after 2022, consistent with a regime characterized by frequent reversals and limited trend persistence in the most liquid energy benchmarks. While the Core-6 momentum portfolio benefited from episodic volatility driven trends, most notably from being short the sharp decline in oil prices during the COVID shock and long the dislocation in Gasoil following the Russia Ukraine war, these gains were largely reversed as markets transitioned into a more range bound environment. By late 2024, cumulative performance had retraced to approximately breakeven relative to the start of the sample in 2015, turning negative thereafter. This pattern underscores the fragility of short horizon momentum in mature and highly crowded markets.

By contrast, momentum in NGL futures displays a more persistent and stable return profile over the full sample. Although performance moderates after 2022, the NGL momentum portfolio avoids the prolonged drawdowns observed in the Core-6 and retains a positive cumulative trajectory. This behavior is consistent with stronger structural trends and lower crowding in newly liquid markets, where evolving market participation and infrastructure allow trends to develop over longer horizons.

When viewed jointly, these results suggest that NGL momentum provides meaningful diversification relative to classical energy momentum. Extending trend following exposure beyond the most liquid benchmarks allows the portfolio to regain access to persistent price dynamics that historically underpinned momentum premia in commodity markets. Consistent with this interpretation, the equally weighted momentum portfolio exhibits dynamics similar to the Core-6 but benefits from the inclusion of NGLs during periods when traditional energy momentum struggles.

More broadly, these findings reinforce a central insight of the momentum literature. Diversification across a wider cross section of markets is critical to sustaining trend following performance. Rather than relying solely on established benchmarks, incorporating newly liquid contracts can restore exposure to emerging trends and mitigate the limitations imposed by crowding and regime dependence.

5.2 Carry: Results

At first glance, the carry results in Table 2 suggest a robust and stable risk premia within the Core-6 universe. Returns and information ratios remain positive across subsamples, and one might conclude that carry continues to perform well even in the post-2022 environment. Statistically, this interpretation is not wrong. However, examining the time-series behavior of the strategy reveals a more nuanced picture.

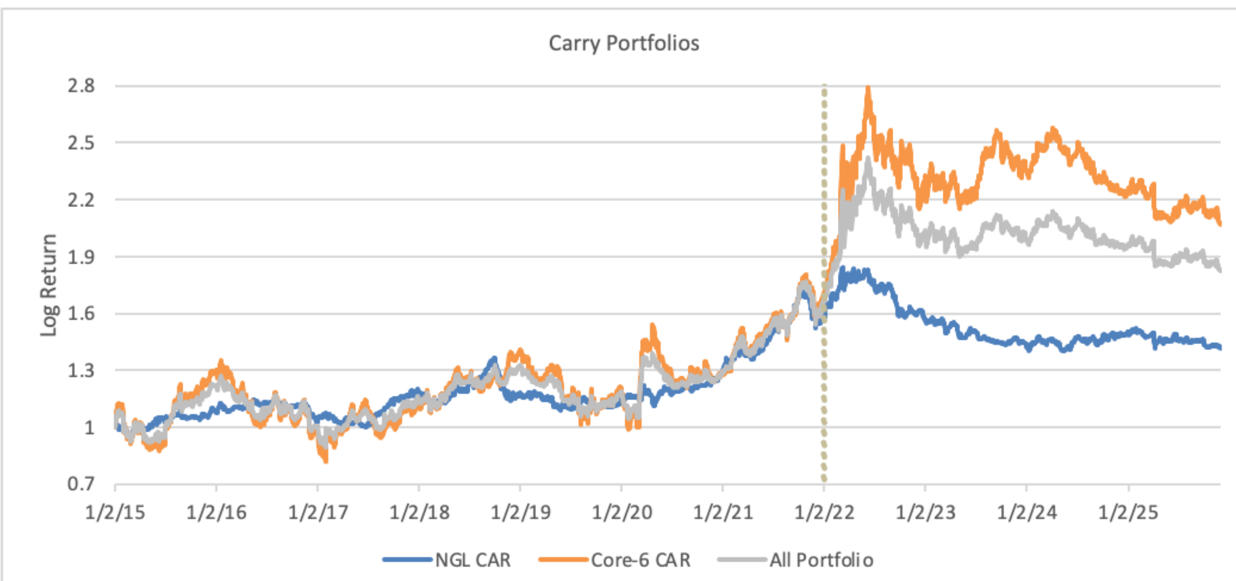

Figure 2 Carry performance for equally weighted portfolios of Core-6 energy futures and NGL futures over the full sample period. The "All Portfolio" series represents an equally weighted carry portfolio across all nine commodities. The vertical dotted line marks January 1, 2022.

Figure 2 shows that the Core-6 carry portfolio experienced an exceptional run from 2020 through mid-2022, coinciding with pronounced dislocations in global energy markets. The strategy was aligned with these dislocations, producing unusually strong and persistent carry returns. As a result, the January 2, 2022, sample split places much of this outperformance in the post-2022 subsample, overstating the robustness of carry in summary statistics. Outside this episode, performance has been more uneven. Although the strategy has avoided sustained losses in recent years, returns have become increasingly choppy, reflecting more irregular term-structure dynamics and weaker persistence of traditional carry signals.

These dynamics are consistent with a market environment in which market forces generate sharp but short-lived curve distortions, rather than the stable contango or backwardation regimes that historically supported carry premia. As a result, carry performance increasingly depends on infrequent but extreme episodes rather than steady accumulation over time.

In NGL markets, the picture changes sharply, with carry returns substantially weaker. Unlike crude oil and distillates, NGL futures curves are not dominated by traditional inventory hedgers and exhibit less persistent term-structure signals. Instead, NGL term structures are more heavily influenced by episodic fractionation constraints and regional bottlenecks, which limits the durability of carry signals. As shown in Figure 2, NGL carry tracks Core-6 carry reasonably closely through 2021, but its recent performance deteriorates as these structural features become more binding.

As a result, NGL carry contributes little standalone value over the full sample. When combined with the Core-6, its inclusion does not materially impair portfolio performance, reflecting the dominant role of mature energy markets in determining aggregate carry returns. These results reinforce the interpretation of carry as a risk premium closely linked to physical storage economics, which remain intermittently powerful in established markets but are less reliably expressed in newer and more fragmented NGL systems.

5.3 Value: Results

We begin by examining the performance of the standalone value signal. As shown in Table 2, value delivers positive but modest returns across subsamples, with moderate volatility and information ratios relative to the other strategies. While these summary statistics provide a useful baseline, they mask substantial time variation in performance and do not fully capture the interaction between value and other signals. To better understand the role of value within the broader strategy set, we therefore present the equity lines of momentum, value, and their composite signal for the full nine-commodity universe across the split sample period. Although the composite momentum–value signal is not included as a separate sleeve in the final portfolio to avoid double counting as we include the strategies separately, its blended performance remains economically meaningful and is therefore analyzed explicitly. Additionally, we provide equity curves for standalone value applied to the Core-6, NGLs, and an equally weighted portfolio are provided in Appendix G.

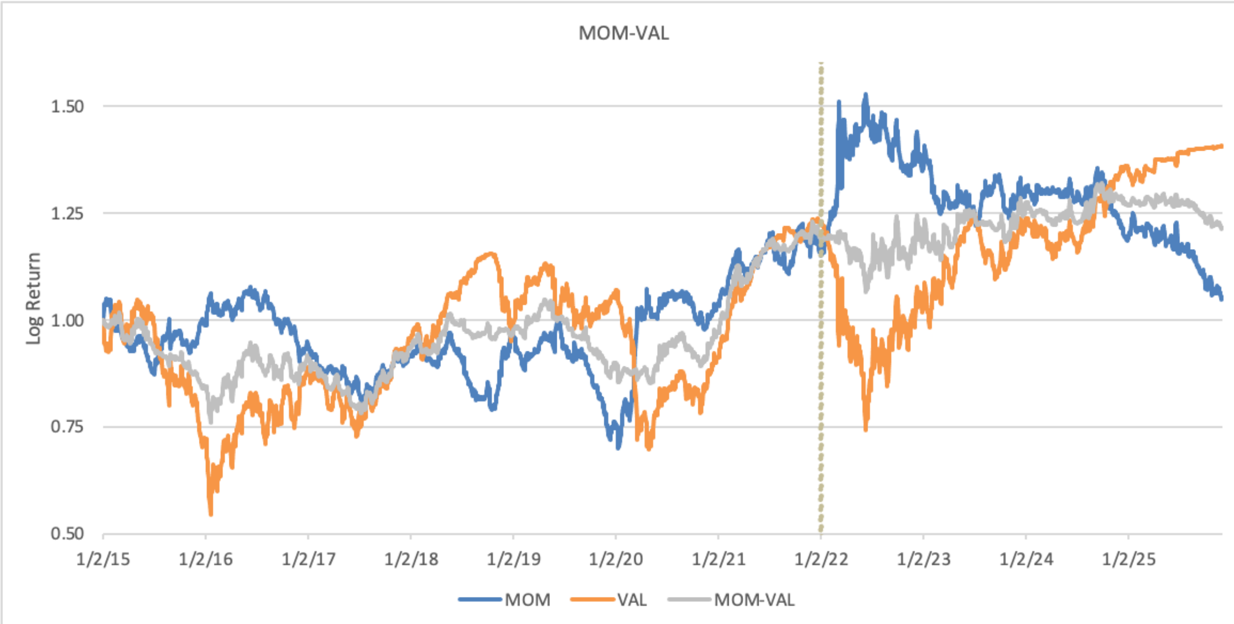

Figure 3 Cumulative log return performance of momentum, value, and their composite momentum–value signal for an equally weighted portfolio of all nine commodities. The vertical dotted line marks January 1, 2022.

Figure 3 highlights the cyclical nature of value performance, characterized by extended drawdowns followed by periods of recovery rather than prolonged inactivity. Value contributes meaningfully during several distinct episodes, including 2016–2019, much of 2020 through the end of 2021, and again beginning in late 2022. The 2022 sample split occurs near a local peak in value performance, mechanically depressing post-split averages even as value subsequently resumes contributing and reaches new highs as market conditions evolve.

These cycles include relatively sharp drawdowns that materially influence summary statistics without fully erasing prior gains. Such drawdowns tend to coincide with periods in which prices move persistently away from historical levels following large global shocks, reflecting episodes in which the value signal calls for reversion while the underlying market has undergone a regime shift. As a result, the modest average performance of value in Table 2 reflects timing effects rather than a persistent lack of efficacy.

The interaction between momentum and value further clarifies that the two signals are not symmetric opposites. While they frequently point in opposing directions, their joint deployment generates positive returns with substantially lower volatility than either signal in isolation, as shown in both Figure 3 and Table 2. This outcome does not arise from mechanical cancellation. Momentum and value are implemented on different maturities of the futures curve, with momentum expressed in shorter dated contracts and value positioned further out on the term structure and activated only during sufficiently large deviations through a wide threshold band. As a result, opposing signals do not eliminate exposure, and the combined signal retains meaningful risk and return.

Running both signals simultaneously produces a more hedged return profile that remains engaged across regimes. During periods when trend persistence weakens and momentum experiences drawdowns, value tends to recover as prices revert toward longer horizon reference levels. Conversely, when strong trends dominate, momentum drives performance while value exposure is limited by its activation threshold. The resulting interaction improves robustness without relying on either signal to perform continuously.

Taken together, these results illustrate that standalone backtests can provide an incomplete picture of systematic strategy behavior. Signals grounded in economic mechanisms may experience extended drawdowns or long periods of weak performance before becoming relevant under changing market conditions. Evaluating such signals solely on the basis of isolated performance windows risks overlooking their role in stabilizing returns across regimes. In this sense, the momentum–value interaction serves less as evidence of a new standalone premia than as a demonstration of how economically motivated signals can complement one another in practice.

From a portfolio construction perspective, these findings suggest that evaluating momentum and value in isolation can understate their contribution within a systematic framework. While each signal may appear fragile when considered alone, their joint deployment produces a more robust return profile by diversifying exposure across time horizons. The momentum–value interaction therefore illustrates how combining signals with distinct economic foundations can improve stability and resilience in systematic energy portfolios, reinforcing the importance of diversification across horizons rather than reliance on any single source of return.

5.4 Statistical Arbitrage: Results

We next present results for the eight selected statistical arbitrage spreads, showing that strong portfolio-level performance arises primarily from low cross-spread correlations rather than uniformly strong standalone returns.

STAT-ARB	Ethane-Natgas	Propane-Ethane	Propane-Butane	RBOB-Butane	RBOB-Brent	Brent-ULSD	ULSD-WTI	WTI-Brent
Ethane-Natgas		0.05	0.02	0.03	0.02	0.00	0.00	0.00
Propane-Ethane	0.05		-0.07	-0.18	0.03	-0.05	-0.06	-0.01
Propane-Butane	0.02	-0.07		0.06	-0.06	-0.06	-0.03	0.00
RBOB-Butane	0.03	-0.18	0.06		0.16	-0.06	-0.04	0.09
RBOB-Brent	0.02	0.03	-0.06	0.16		0.06	0.04	-0.01
Brent-ULSD	0.00	-0.05	-0.06	-0.06	0.06		0.70	-0.08
ULSD-WTI	0.00	-0.06	-0.03	-0.04	0.04	0.70		0.26
WTI-Brent	0.00	-0.01	0.00	0.09	-0.01	-0.08	0.26	
Return	10%	15%	1%	14%	6%	0%	4%	6%
Vol	20%	44%	20%	31%	8%	10%	13%	9%
IR	0.51	0.33	0.05	0.45	0.76	-0.05	0.29	0.65

Table 3: Correlations among eight chosen spreads (upper panel), with spread performance reported (lower panel)

Table 3 shows correlations across most statistical arbitrage spreads are close to zero or mildly negative, with the notable exceptions of ULSD–WTI and Brent–ULSD. These higher correlations are largely mechanical, reflecting the fact that Brent and WTI themselves are highly correlated, so the associated spreads effectively represent closely related expressions of the same underlying price dynamics. Outside of these cases, near-zero cross-spread correlations provide substantial diversification benefits. As shown later, the statistical arbitrage portfolio is also weakly correlated with other risk premia, reinforcing the role of statistical arbitrage as a diversifying strategy within the broader portfolio.

Beyond diversification, the majority of spreads exhibit relatively strong risk-adjusted performance. Most pairs generate positive returns, and several deliver particularly attractive information ratios. RBOB-Brent stands out as one of the strongest performers, while NGL-based spreads also contribute meaningfully, with a larger share of their performance concentrated in recent years. For example, Ethane–Natural Gas achieves an information ratio close to 1.00 over the most recent sample period. Full split-sample statistics and individual equity curves for all spreads are reported in Appendix I.

Performance nevertheless varies meaningfully across spreads, underscoring that statistical arbitrage is not a one-size-fits-all strategy. ULSD-linked spreads, for example, deteriorate sharply after 2022, coinciding with persistent dislocations and elevated volatility near the front end of the ULSD futures curve. Although positions are implemented using a year-out end-of-month roll, fair value is estimated using a relatively short 20-day moving average, which may be insufficient for spreads exposed to structural or regime-driven shifts. More broadly, this heterogeneity highlights the importance of spread-specific specification and ongoing monitoring, and helps explain why statistical arbitrage is often managed as a specialized product rather than a fully generic signal.

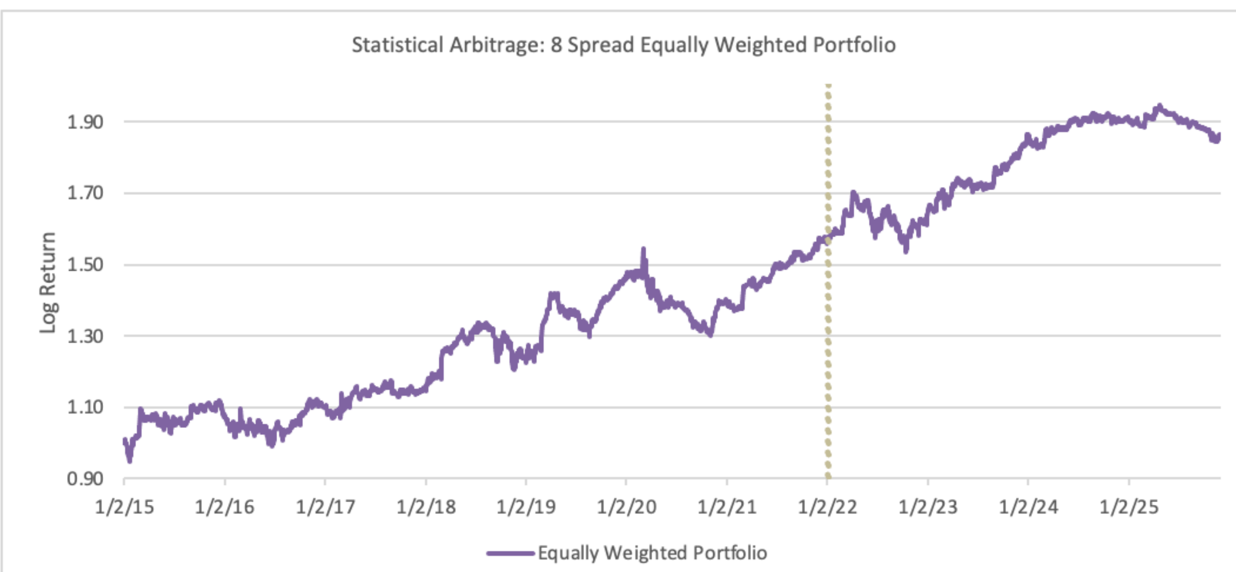

Figure 4 Log-return equity curve of an equally weighted portfolio of eight statistical arbitrage spreads, illustrating the benefits of diversification across largely uncorrelated relative-value trades

Figure 4 presents the equally weighted statistical arbitrage portfolio formed from the eight spread strategies. Combining largely uncorrelated relative-value trades produces a smooth equity curve with less pronounced drawdowns than the other strategies considered. Performance remains robust across both subsamples, illustrating the diversification benefits inherent in aggregating economically distinct spread trades.

A defining challenge of statistical arbitrage is determining when the signal should be active. The strategy involves a fundamental tradeoff between insufficient responsiveness and excessive parameter tuning and is therefore inherently exposed to tail risk. Losses tend to arise when mean-reversion signals remain engaged during persistent regime shifts rather than transient dislocations, leading to extended drawdowns before relationships re-anchor. In this respect, statistical arbitrage closely resembles value strategies, which also perform poorly when prices move persistently away from historical anchors following structural change.

For this reason, we emphasize that the implementation here benefits from alignment between strategy inception and the initial emergence of certain economically relevant linkages within the sample. This introduces a modest timing bias, as some spreads are effectively traded from the onset of regimes that only became apparent during the sample period. Accordingly, our focus in what follows is on portfolio-level behavior rather than precise spread-level timing.

5.5 Correlation Structure and Portfolio Integration

In light of the diversification benefits observed within the statistical arbitrage sleeve, we now examine correlations at the strategy level. Table 5 reports pairwise correlations across the five core strategies over the 2015–2025 sample period. The results highlight multiple opportunities for diversification, with several strategies exhibiting near-zero or negative correlations, supporting their joint inclusion within a multi-strategy portfolio.

Strategies	NGL MOM	CORE-6 MOM	CORE-6 CAR	ALL 9 VAL	STAT ARB
NGL MOM		0.36	0.12	-0.09	-0.11
CORE-6 MOM	0.36		0.25	-0.18	-0.14
CORE-6 CAR	0.12	0.25		-0.58	-0.10
ALL 9 VAL	-0.09	-0.18	-0.58		0.22
STAT ARB PORT	-0.11	-0.14	-0.10	0.22	

Table 4 Pairwise correlations across strategies over the full sample period (2015-2025). Low and negative correlations across several strategy pairs indicate substantial diversification potential at the portfolio level. Correlation matrices for individual commodities within each strategy are reported in Appendix I.

Taken together, these correlations translate into strong portfolio-level risk-adjusted performance. As shown in Table 2, combining the five strategies yields an information ratio close to 1 under simple equal weighting, despite meaningful heterogeneity in standalone strategy performance. The resulting return profile is smooth and is driven primarily by diversification rather than by any single dominant return source. This finding reinforces a central theme of this study: robustness in systematic energy portfolios arises from combining economically distinct premia rather than from fine-tuning individual signals.

Importantly, these results are achieved without any explicit strategy-level optimization. Relatively low and stable cross-strategy correlations allow even a simple equally weighted portfolio to capture substantial diversification benefits. For this reason, we defer a detailed analysis of portfolio construction and weighting schemes to the final section of the paper, where we compare equal weighting with more structured allocation approaches and assess the robustness of portfolio performance under alternative specifications.

6. Shift to Real World Mechanics

Following the presentation of results, we now turn to practical implementation. Up to this point, the discussion has focused on the construction and behavior of individual signals under simplified and controlled assumptions intended to isolate their economic properties. In practice, however, real world deployment requires careful consideration of how exposures are implemented and rolled across the futures curve. Portfolio sizing and allocation are discussed separately, abstracting from implementation frictions in order to focus on the interaction and aggregation of signals.

Each strategy contains numerous dynamic components, and these complexities become more pronounced when the strategies are combined into a unified portfolio. The objective of this section is therefore to bridge theory and practice by identifying the implementation choices that materially shape realized performance and by outlining coherent approaches for addressing these challenges.

6.1 Stylized Facts About Roll Returns

For many energy commodities, the notion of a “spot market” is fundamentally different from other asset classes. Physical spot transactions require the buyer to accept delivery at a specific hub such as Cushing for WTI, Mont Belvieu for NGLs, or New York Harbor for refined products, together with the logistical capacity to handle storage, transportation, and quality specifications. As a result, what is commonly referred to as the “spot price” in practice is the price of the nearest to expiry futures contract. This convention reflects the assumption that delivery would occur through the futures mechanism rather than through an immediate physical exchange.

Due to this structure, true economic exposure to a commodity is achieved by holding the front futures contract and systematically rolling it forward as expiry approaches. Financial traders cannot, and would not want to, take delivery, especially in markets where physical handling, pipeline nomination, and quality tolerances create prohibitive frictions. The realized return to holding a commodity future therefore consists of two components: the price change of the front contract itself and the roll return earned or lost when the position is closed and re-established further along the curve.

The choice of how to perform this roll, referred to as the roll schedule, is a primary implementation decision. Different commodities exhibit distinct liquidity, volatility, seasonal patterns, and expiry specifications across the curve, which makes consistent and efficient rolling difficult to execute. In practice, large CTAs and commodity managers calibrate their roll schedule in conjunction with a risk model, often VaR-based, to control exposure spikes and avoid rolling during periods of stressed liquidity or abnormal volatility.

To evaluate how roll mechanics influence systematic strategy performance, we simulate a range of rolling schemes across commodities. These experiments allow us to isolate how different roll choices translate into differences in realized returns, volatility, and signal quality. Across markets and parameter sets, several clear stylized facts emerge regarding the interaction between roll parameters and systematic signals.

6.1.1 Universal Facts for Single Commodities

The empirical behavior of long futures positions aligns closely with the economic theory of storage. Long rolling positions earn superior returns when the forward curve is predominantly in backwardation and perform worse in contango. As described earlier in the exposition of carry, backwardation embeds positive roll yield that arises when futures prices lie below the spot price, creating upward price pressure as the contract converges toward spot. This serves as a direct empirical validation of storage-based pricing theory.

	WTI	Brent	ULSD	RBOB	Gasoil	Natgas	Propane	Butane	Ethane
Total Roll Return	-	+	+	+	+	-	+	+	-
Backwardation	57%	67%	54%	74%	58%	27%	65%	82%	40%
Contango	43%	33%	46%	26%	42%	73%	35%	18%	60%

Table 5 Total annualized roll returns and term-structure regimes by commodity over 2010–2025. Roll returns generally follow the theory of storage; WTI is the notable exception due to asymmetric roll dynamics driven by extended periods of shallow backwardation punctuated by episodes of extreme contango

Shifting the rolling contract further along the curve reduces return volatility. Deferred maturities sit on flatter and more stable segments of the forward curve, where short-term shocks related to inventories, refinery outages, pipeline flows, or local imbalances have a diminished impact on pricing. Shifting exposure from the front of the futures curve toward more deferred contracts mechanically smooths the return series. The cost of this added stability is reduced sensitivity to short-dated price dynamics, which can attenuate economically relevant information embedded at the front of the curve and disproportionately affect trend-following strategies that rely on timely and volatile price movements.

As a result, rolling further out into the curve also reduces total profit/loss. The smoothing introduced by deferred contracts limits the ability of both trend and term-structure signals to monetize curve dislocations. Although lower volatility may appear beneficial in isolation, it is accompanied by weaker directional swings and smaller carry spreads, both of which compress expected returns. For a pure long position, selecting a more deferred roll schedule therefore lowers the risk-adjusted return even though the return path becomes visually smoother.

The mechanical impact of roll choice also alters the strength of any signal applied to the futures curve. Rolling further into the curve is inherently dilutive: although the signal itself remains unchanged, the amplitude of the underlying price response is muted because deferred contracts experience smaller relative changes. In the flatter portions of the curve, even a strong long or short signal translates into a smaller economic exposure, weakening realized performance.

The dilution effect becomes clearer when plotting roll-adjusted prices across tenors and time. Figure 5 illustrates this mechanism using WTI as an example, showing end-of-month roll-adjusted price series constructed at increasing levels of curve maturity.

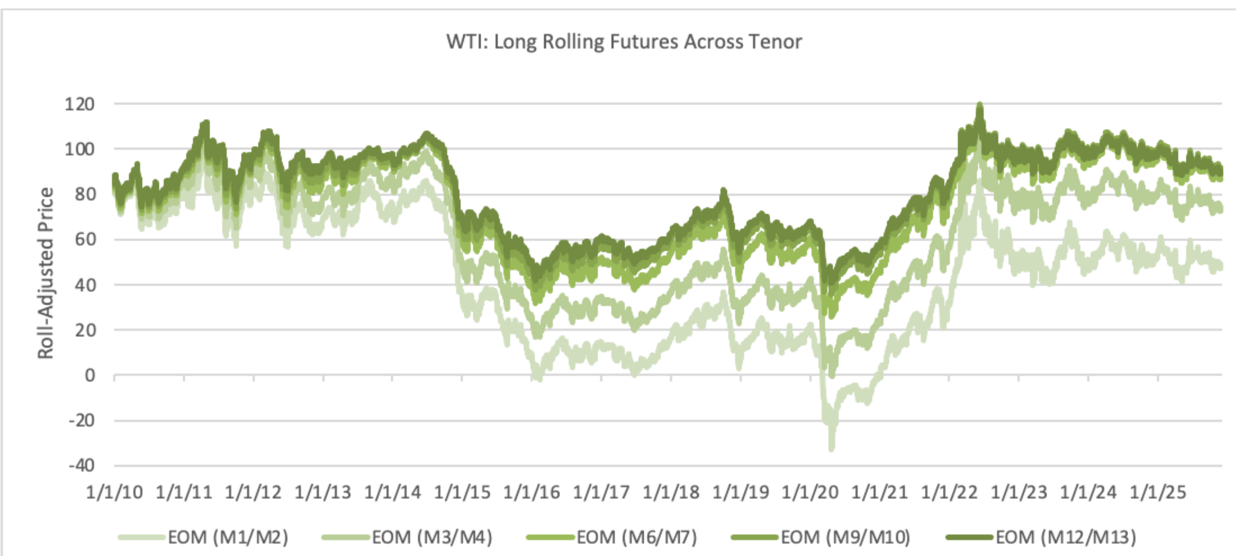

Figure 5 Roll -adjusted WTI price series constructed using end-of-month roll schedules at increasing curve maturities. For each roll pair (e.g., M1/M2, M3/M4, …, M12/M13), the position enters the deferred contract at the start of the month, allows it to age into the front contract over time, and exits near month-end before rolling into the next tenor. Series correspond to progressively more deferred segments of the futures curve.

As roll exposure moves further out along the curve, price paths become visibly smoother and less volatile, consistent with deferred contracts lying on flatter and more stable segments of the forward curve. At the same time, large front-end dislocations such as the

2020 collapse are progressively muted at longer maturities, illustrating how economically relevant shocks are diluted as exposure shifts away from the front of the curve. Finally, the compression of price swings highlights the inherent tradeoff in roll selection: while deferred contracts reduce volatility, they also dampen exposure to the very price dynamics that generate risk-adjusted returns, limiting the ability of systematic strategies to monetize curve movements.

6.1.2 Implications for Risk Premia

The downstream effects of roll selection are asymmetric across risk premia strategies. Increased roll-induced price volatility tends to enhance momentum performance while impairing carry. This asymmetry is intuitive. Momentum strategies benefit from directional price movement and perform best when higher volatility reveals stronger and more persistent short-term trends. Carry strategies, by contrast, rely on the stability of the existing term structure and are effectively bets on persistence. When front-month volatility rises, carry becomes vulnerable to sharp reversals, behaving like an exposure that accumulates small gains while remaining exposed to infrequent but severe losses. We illustrate these dynamics using WTI momentum performance across roll tenors and time in Figure 6.

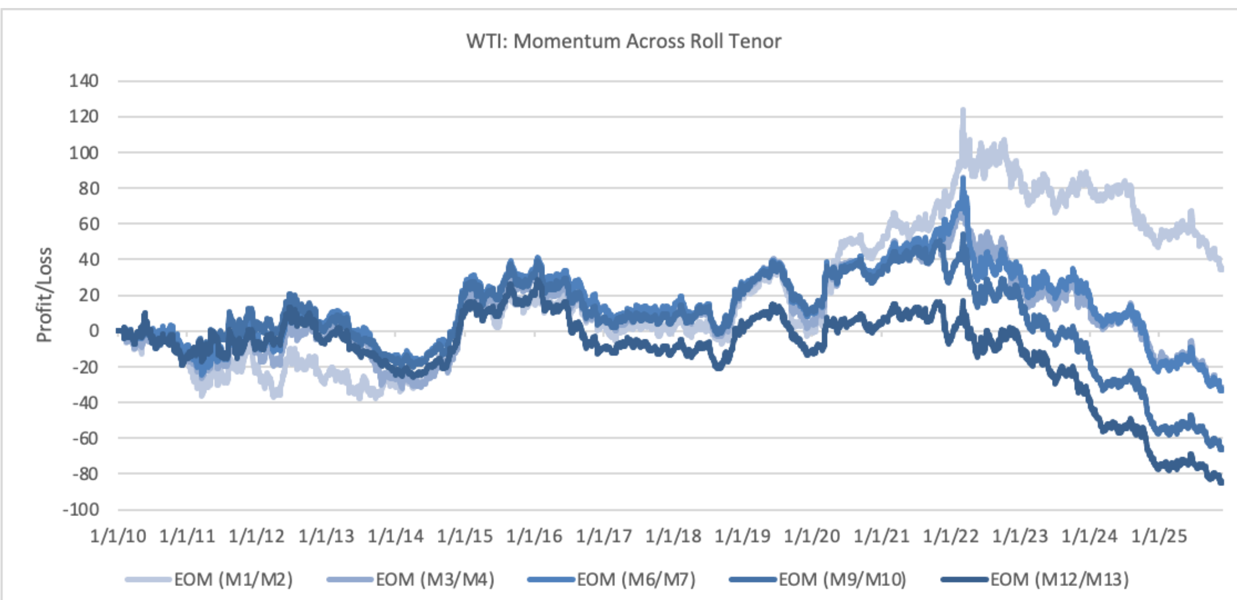

Figure 6 Cumulative profit/loss per barrel for WTI momentum across roll tenors (2010–2025).

Momentum strategies exhibit a pronounced dependence on rolling horizon. Shorter roll schedules consistently deliver superior risk-adjusted returns relative to more deferred alternatives. This pattern aligns with the short-horizon design of moving-average signals, which are intended to capture relatively fast-moving trends. The front of the curve concentrates both volatility and price discovery, generating stronger trend signals. As a result, momentum behaves as an expiration-adjacent strategy: the objective is to remain as close to the front as feasible while avoiding delivery or liquidity risk, thereby maximizing exposure to informative price movements. We next examine the same roll-tenor dependence for carry strategies in Figure 7.

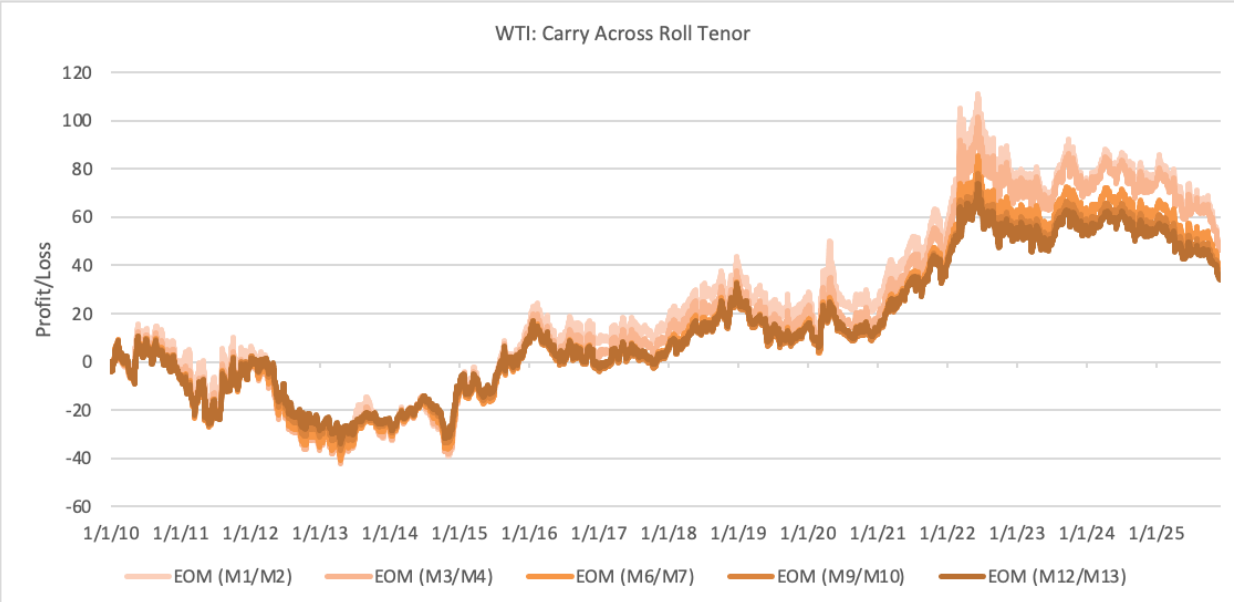

Figure 7 Cumulative profit/loss per barrel for WTI carry across roll tenors (2010–2025).

Carry strategies exhibit the opposite behavior. Longer roll horizons tend to improve performance, consistent with carry being a slow-moving signal driven by persistent spreads across maturities. By shifting exposure away from the front month, the strategy avoids transient curve inversions and tail events that disproportionately harm carry returns. More deferred contracts effectively impose a natural threshold, activating the signal only when the curve is materially steep and reducing exposure to front-month volatility that erodes carry performance. This mechanism can be interpreted as a built-in ε filter on curve steepness.

For value and statistical arbitrage strategies, roll horizon is not a freely optimizable design parameter in the same way as momentum and carry. These strategies depend on deviations from long-run equilibrium relationships that are specific to each commodity or cross-commodity spread. Value signals rarely exhibit short-term efficacy, as they often trade against contemporaneous roll yield, requiring more deferred contracts and wider thresholds to ensure that only persistent mispricings are targeted. Statistical arbitrage strategies rely on economically grounded relationships within a given pair or spread, and mechanically varying the roll horizon would distort the intended exposure implied by the trading thesis. For these mean-reversion strategies, roll choice is therefore dictated by theoretical coherence and empirical stability rather than by choice.

6.2 Sizing a Portfolio of Systematic Strategies

In practice, a portfolio of systematic signals is only useful to the extent that it can be translated into actionable positions. Individual signals may issue conflicting recommendations. For example, momentum may indicate a long exposure while value simultaneously suggests a short. A strategy portfolio often contains many of these signals with conflicting micro-views. The practical question is which signals to act on and how much of each commodity to buy or sell.

A natural idea from a quantitative perspective is to rely on a numerical optimizer to choose the combination of strategies that maximizes the in-sample Sharpe ratio. While appealing in theory, this approach is extremely sensitive to estimation error. Optimizing directly on Sharpe amplifies noise in both expected returns and covariances, often producing unstable allocations. Some styles, especially momentum, are particularly vulnerable in high-dimensional settings, where small input changes lead to large weight fluctuations.

For these reasons, we begin with the most basic ingredient: *what do the signals themselves say?*

6.2.1 Identifying Signal Based Exposures

Each composite strategy produces a normalized signal in the range [−1, +1], where −1 denotes a full short position, +1 a full long, and 0 a neutral stance. Collecting these signals across all strategies at time (t) gives the signal vector:

$$\boldsymbol{S}_t = \begin{pmatrix} s_{1,t} \\ s_{2,t} \\ \vdots \\ s_{M,t} \end{pmatrix}, \qquad s_{i,t} \in [-1,1].$$

Where M denotes the number of strategies.

On their own, these signals contain no information about position size. They represent only the directional view implied by each model at time. To convert these views into tradable exposures, we scale each signal by the capital allocated to the trading system.
Let C denote this capital. Then the notional exposure assigned to strategy j is

$$E_{j,t} = C \cdot s_{j,t}.$$

Repeating this for all strategies yields the strategy-exposure vector, which specifies the notional capital allocated to each strategy:

$$\boldsymbol{E}_t = C\boldsymbol{S}_t = \begin{pmatrix} Cs_{1,t} \\ Cs_{2,t} \\ \vdots \\ Cs_{M,t} \end{pmatrix}.$$

Each strategy distributes its exposure across the underlying commodities that it trades. Let N denote the number of assets. Define $B_{i,j,t}$ as the fraction of exposure strategy j that is allocated to asset i at time t. Collecting these values yields the asset-strategy exposure matrix

$$\boldsymbol{B}_t = \begin{bmatrix} B_{1,1,t} & B_{1,2,t} & \cdots & B_{1,M,t} \\ B_{2,1,t} & B_{2,2,t} & \cdots & B_{2,M,t} \\ \vdots & \vdots & \ddots & \vdots \\ B_{N,1,t} & B_{N,2,t} & \cdots & B_{N,M,t} \end{bmatrix}.$$

Multiplying the strategy exposure vector by this mapping yields the corresponding asset-level exposures

$$\boldsymbol{X}_t = \boldsymbol{B}_t \boldsymbol{E}_t = \begin{pmatrix} X_{1,t} \\ X_{2,t} \\ \vdots \\ X_{N,t} \end{pmatrix}.$$

The vector $\boldsymbol{X}_t$ represents the final dollar exposure to each underlying commodity implied by all strategies at time t. It is, in effect, the trade recommendation produced by the signal engine: a money-weighted aggregation of directional views, collapsed from strategy space into commodity space. At this stage, we make an important design choice not to optimize or reweight exposures further. Incorporating strategy-level return dynamics into an optimizer can introduce substantial noise, and systematic signals are known to behave poorly when directly optimized. We therefore separate signal formation from risk allocation, deferring all optimization decisions to the asset level in the next section.

While the mapping above is expressed in dollar space to reflect real-world implementation, our empirical analysis is conducted using normalized log returns, under which portfolio performance is invariant to uniform scaling of exposure. As a result, we can work directly with the relative asset-level exposure structure summarized by $\boldsymbol{B}_t$, without the need for explicit cash investment definitions. The cash-based construction is retained here for completeness and economic interpretation.

We note that the raw exposure vector $\boldsymbol{X}_t$ does not account for differences in volatility or co-movement among the underlying commodities. In its unmanaged form, it may therefore be overly concentrated in a small subset of assets or exhibit excessive variability through time. This motivates the introduction of a covariance-aware risk model and allocation framework, which we develop in the following sections.

6.2.2 Dynamic Risk Model: "Smart" Covariance Matrix

To translate the raw asset-level exposure vector $\boldsymbol{X}_t$ into a coherent measure of portfolio risk, we introduce a dynamic variance–covariance model defined at the commodity level. The objective of this model is to produce risk estimates that are stable, interpretable, and comparable across market conditions, while remaining responsive to evolving volatility and dependence structures. The model captures three elements of the return distribution: individual asset volatilities, cross-asset covariances, and a shrinkage adjustment that stabilizes estimation error. Together, these components yield a well-conditioned covariance matrix suitable for dynamic risk allocation.

We estimate conditional volatilities using an exponentially weighted moving average (EWMA) specification with a 20-day half-life. Commodity markets are prone to abrupt volatility spikes driven by supply disruptions, policy shocks, and rapid repricing of forward curves; as a result, recent returns must receive greater weight. The half-life parameter governs the rate at which historical observations lose influence: a return observed 60 trading days ago receives half the weight of a contemporaneous return, with influence decaying smoothly thereafter. For commodity (i) and its daily log return $\left(r_{i,t}\right)$, the variance can be expressed as:

$$\sigma_{i,t}^2 = (1-\lambda_{\mathrm{vol}})\sum_{\ell=1}^{\infty}\lambda_{\mathrm{vol}}^{\ell-1} r_{i,t-\ell}^2\ , \quad with\ \lambda_{vol} = 2^{(-1/20)}$$

In contrast, correlations among commodities evolve more slowly. Empirically, cross-commodity relationships are “sticky”: correlations products shift gradually and typically do not reverse meaningfully on short horizons. Overreacting to transitory correlation movements can lead to unstable or oscillating risk estimates, with the covariance matrix “whipsawing” the allocation process. To reflect this persistence, we estimate covariances using a longer 60-day half-life:

$$\Sigma_{i,j,t} = (1-\lambda_{\mathrm{cov}})\sum_{\ell=1}^{\infty}\lambda_{\mathrm{cov}}^{\ell-1}(r_{i,t-\ell}r_{j,t-\ell}), \qquad with\ \lambda_{cov} = 2^{(-1/60)}$$

Although the exponentially weighted estimator produces a more stable covariance structure, the raw sample correlations remain highly noisy, particularly in a multi-asset futures universe where effective sample sizes are limited and return distributions exhibit heteroskedasticity. When such matrices are fed into portfolio optimizers, they often induce extreme and economically implausible long/short configurations. This instability arises because optimizers overreact to spurious correlations: small, transient shifts in the sample correlation matrix can translate into disproportionately large changes in optimal weights. As discussed earlier, trend-following systems are especially sensitive to this problem because their exposures fluctuate frequently; without correction, an unstable covariance estimate can therefore generate excessive turnover and unintended concentration risk.

To mitigate these effects, we apply a diagonal shrinkage transformation of the Ledoit–Wolf form, pulling correlations toward zero and tempering the influence of noisy off-diagonal elements. The resulting estimator blends the empirical covariance matrix with its diagonal:

$$\bar{\Sigma_t} = (1-\alpha) * \Sigma_t + \alpha * diag(\Sigma_t), \qquad with\ \alpha = 0.5$$

where α controls the degree of shrinkage. This procedure reduces estimation error, improves the conditioning of the matrix, and prevents the optimizer from amplifying statistical artifacts. It yields a more robust representation of the joint risk structure, ensuring that allocations respond to genuine changes in market conditions rather than to random fluctuations in estimated correlations.

The shrunk covariance matrix $\bar{\Sigma_t}$ thus serves as the core of the risk model. It provides a stable mapping from the exposure vector $\boldsymbol{X_t}$ to the system's overall risk footprint, forming the basis for the risk adjusted dynamic allocation mechanism described in the next section.

6.2.3 Risk Parity Allocation

Having constructed a time series of rolling realized volatility estimates and a rolling shrunk covariance matrix at the asset level, we now combine these elements into a portfolio-level allocation via a risk-parity weighting scheme. At this stage, all strategy-level signals have already been aggregated into a single vector of asset-level exposures, and risk allocation is performed across underlying commodities rather than across strategies.

The objective is to convert the system's raw asset exposures into a set of asset level exposures (portfolio weights) that reflect each asset's inherent volatility profile while maintaining a consistent level of total portfolio risk through time. We begin by rescaling each asset's raw exposure ($X_{i,t}$) by the inverse of its estimated 20-day volatility ($\sigma_{i,t}$). This produces a vector of unnormalized risk-parity weights

$$\omega_{0,i,t} = \frac{X_{i,t}}{\sigma_{i,t}},$$

which penalizes high-volatility assets and expands the influence of lower-volatility ones. This inverse-volatility transformation is the core of the risk-parity principle: it ignores contemporaneous correlations and equalizes the standalone risk of each component using the most stable and reliably estimated part of the risk model. The unscaled vector ($\omega_{0,t}$) implies a portfolio volatility under the shrunk covariance matrix ($\widetilde{\Sigma_t}$), which we compute as

$$\sigma_{p,t} = \sqrt{\omega_{0,t}^{\top} \widetilde{\Sigma_t \omega_{0,t}}}.$$

To maintain a consistent level of portfolio risk through time, we rescale the entire exposure vector by a time-varying scalar (γ_t) chosen to match a desired volatility target (σ_{target}):

$$\gamma_t = \frac{\sigma_{\text{target}}}{\sigma_{p,t}}, \qquad \omega_t = \gamma_t \omega_{0,t}.$$

This adjustment affects only the overall size of the portfolio while preserving the relative structure implied by the inverse-volatility step. The final implemented exposures can be written succinctly as

$$\widetilde{X_t} = \gamma_t \left(\frac{X_t}{\sigma_t}\right),$$

where the inverse volatility is applied elementwise. For interpretability purposes, these weights may be normalized to sum to one; however, such normalization has no effect on their underlying risk properties or the portfolio's volatility.

Risk parity, combined with our rolling risk estimation, provides a transparent and robust framework for aggregating heterogeneous strategies into a unified, risk-controlled portfolio. It avoids the instability and turnover associated with full covariance-based optimizers while ensuring that no individual asset dominates the aggregate risk of the system. Moreover, because the procedure is applied to a rolling time series of volatility and covariance estimates, the resulting weights adapt naturally to evolving market conditions, expanding or contracting as variance and co-movements shift over time.

6.2.4 Comparison with Equal Weights

As a final step, we evaluate the performance and risk characteristics of the dynamic risk parity allocation relative to an equal weighted benchmark. This comparison isolates the contribution of volatility scaling to overall portfolio behavior and provides a reference point for interpreting the effects of the risk model. Table 6 reports split sample performance statistics for portfolios constructed under equal weighting and dynamic risk parity.

Strategy Portfolios		**Full Sample**	**2015-2022**	**2022-2025**
EW PORT	Return	4%	5%	3%
	Vol	5%	5%	5%
	IR	0.91	1.03	0.66
DRP PORT	Return	5%	6%	4%
	Vol	5%	6%	3%
	IR	1.03	1.05	1.09

Table 6 Split sample statistics for the equally weighted (EW) portfolio of systematic strategies and the dynamic risk parity (DRP) implementation.

Across the full sample, the dynamic risk parity portfolio delivers a modestly higher average return while maintaining a similar level of realized volatility, resulting in an improvement in risk adjusted performance relative to the equal weighted benchmark. More importantly, the improvement in information ratios is persistent across subsamples. In particular, dynamic risk parity exhibits greater stability in the post 2022 period, despite heightened volatility and weaker performance in several underlying strategy components. We see it delivering higher return at lower volatility compared to the equal weighting scheme.

These results are consistent with the design of the asset level volatility scaling mechanism. By systematically reducing exposure to assets experiencing elevated volatility, the dynamic risk parity framework limits the transmission of localized risk shocks to the aggregate portfolio. As a result, portfolio risk remains more evenly distributed through time, leading to smoother performance and more consistent risk adjusted returns without relying on full covariance-based optimization.

Figure 8 illustrates these dynamics by comparing the cumulative log equity curves of the equal weighted and dynamic risk parity portfolios over the full sample period. While both portfolios follow similar long run trends, the dynamic risk parity portfolio exhibits reduced drawdowns during periods of elevated market stress and a smoother trajectory overall, consistent with the improvements in risk adjusted performance reported in Table 6.

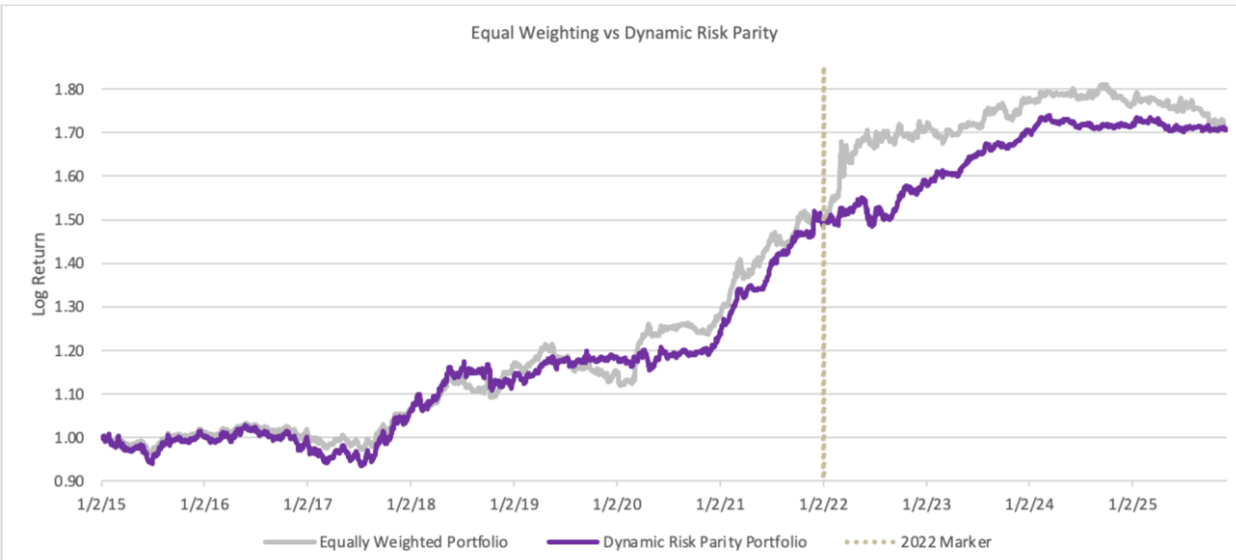

Figure 8 Split sample statistics for the equally weighted (EW) portfolio of systematic strategies and the dynamic risk parity (DRP) implementation. Volatility is matched at over the full sample for equal comparison.

The differences between the two equity curves are intentionally modest. Because both portfolios are constructed to operate at comparable levels of total risk, the effects of dynamic risk parity appear primarily in volatility dynamics rather than in large divergences in cumulative returns. As a result, the comparison is best understood in terms of return smoothness and drawdown behavior rather than absolute performance.

The dynamic risk parity portfolio exhibits a smoother return path, with generally shallower and less abrupt drawdowns during periods of elevated market volatility. These differences arise mechanically from the asset level volatility scaling, which reduces exposure when individual assets or strategy components become more volatile and limits the propagation of localized shocks to the aggregate portfolio.

From an implementation standpoint, this distinction is economically relevant. Tighter volatility control allows the portfolio to be levered more reliably within a risk constrained framework. When evaluated on a risk equivalent basis, the improvements in risk adjusted performance documented earlier translate directly into higher expected returns, even though unlevered equity curves remain close.

Overall, the results highlight dynamic risk parity as a risk efficiency mechanism rather than a source of standalone return enhancement. Its primary contribution lies in stabilizing portfolio behavior and improving the consistency of risk deployment through time, particularly during periods of heightened volatility.

7. Conclusion

This study reexamines the behavior of classical risk premia in commodity futures and investigates how their performance has evolved in modern market conditions. By revisiting the signals that historically defined systematic commodity investing, we show that the apparent weakening of traditional premia cannot be attributed solely to overcrowding or permanent structural change. Instead, a meaningful share of the opportunity appears to lie in broadening the investment universe to include less heavily traded commodities, incorporating economically motivated “quantamental” information, and revisiting established signals whose relevance may reemerge under changing market regimes.

Throughout the analysis, we focus on what is realistically implementable. This requires addressing the practical constraints that separate an appealing backtest from a tradeable strategy, including liquidity, roll mechanics, tenor selection, and the stability of multivariate risk estimates. Many of these considerations are difficult to capture cleanly in historical simulations, and our results highlight the sensitivity of commodity risk premia to assumptions that are often implicit in empirical studies.

A central contribution of the paper is a time series aware, holdings-based risk framework that maps signal level exposures directly to portfolio level outcomes. By operating on exposures rather than realized returns, the framework avoids look ahead bias in covariance estimation and provides a transparent, adaptive mechanism for combining heterogeneous signals under a unified risk budget. This approach offers a practical blueprint for constructing systematic commodity portfolios that are coherent, risk controlled and grounded in economically interpretable principles.

While no framework can eliminate uncertainty, the tools and diagnostics developed here help clarify where commodity alpha persists, how it has shifted over time, and what is required to capture it in a modern trading environment.

References:

[1] Asness, C., Moskowitz, T., & Pedersen, L. (2013). *Value and Momentum Everywhere*. Journal of Finance, 68(3), 929–985.
https://ssrn.com/abstract=2174501

[2] Baz, J., Granger, N., Harvey, C. R., Le Roux, N., & Rattray, S. (2015). *Dissecting Investment Strategies in the Cross Section and Time Series*. SSRN Working Paper No. 2695101.
https://ssrn.com/abstract=2695101

[3] Bouchouev, I. (2023). *Virtual Barrels: Quantitative Trading in the Oil Market*. Springer.
https://link.springer.com/book/10.1007/978-3-031-36151-7

[4] Bouchouev, I., & Sun, W.-Y. (2025). *Myths and Mysteries About Speculation in the Oil Market*. KAPSARC.
https://www.kapsarc.org/our-offerings/publications/myths-and-mysteries-about-speculation-in-the-oil-market

[5] Bouchouev, I., & Zuo, L. (2020). *Oil Risk Premia under Changing Regimes*. Global Commodities Applied Research Digest, Winter 2020.
http://www.jpmcc-gcard.com/digest-issues/15

[6] Fernandez-Perez, A., Fuertes, A.-M., & Miffre, J. (2021). *The Risk Premia of Energy Futures*. Energy Economics, 102, 105121.
https://www.sciencedirect.com/science/article/abs/pii/S0140988321003467

[7] Koijen, R. S., Moskowitz, T. J., Pedersen, L. H., & Vrugt, E. B. (2018). *Carry*. Journal of Financial Economics, 127(2), 197–225.
https://doi.org/10.1016/j.jfineco.2017.11.004

[8] Ledoit, O., & Wolf, M. (2004). *Honey, I Shrunk the Sample Covariance Matrix*. Journal of Portfolio Management, 30(4), 110–119.
http://www.ledoit.net/honey.pdf

[9] U.S. Energy Information Administration (EIA). (n.d.). *Petroleum & Other Liquids Data*. U.S. Department of Energy.
https://www.eia.gov/petroleum/data.php

Appendix A: Momentum Parameter n Instability

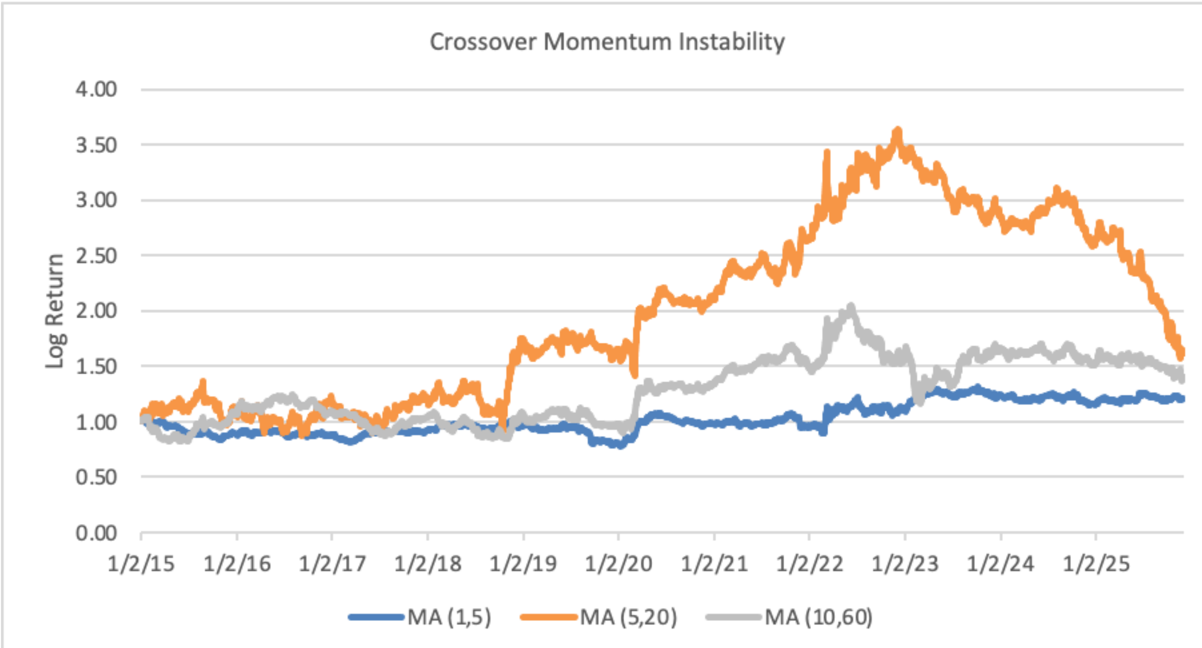

Figure 9 *Cumulative log equity curves for crossover momentum strategies under alternative moving average parameterizations. The figure illustrates the sensitivity of momentum performance to the choice of short and long lookback windows over the full sample period*

Appendix B: Natural Gas Liquid Volume on ICE and CME

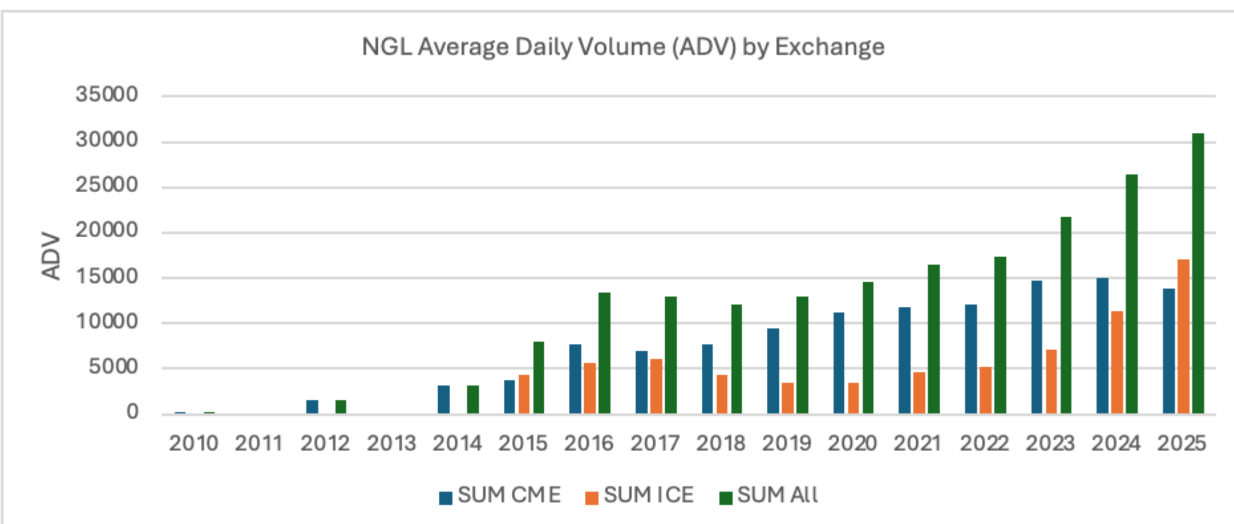

Figure 10 *Average daily trading volume for natural gas liquid futures on CME and ICE exchanges by year. Volumes are aggregated across contracts, highlighting the growth and evolving distribution of liquidity across venues.*

Appendix C: Statistical Arbitrage Economic Rationale

- **Ethane-Natgas:** Linked by extraction and rejection economics, governed by gas processing and fractionation constraints.
- **Propane-Ethane:** Linked by petrochemical feedstock optionality, as propane and ethane compete as steam-cracker inputs
- **Propane-Butane:** Linked by LPG blending and petrochemical feedstock substitution, with relative pricing constrained by switching economics.
- **RBOB-Butane:** Linked by gasoline blending economics, as butane is a marginal blending component governed by vapor-pressure constraints.
- **RBOB-Brent:** Linked by gasoline refining economics, connecting crude input costs to gasoline output via refinery yields and product arbitrage.
- **Brent-ULSD:** Linked by Atlantic Basin refining margins, with spreads anchored by Brent-priced crude inputs and distillate demand.
- **ULSD-WTI:** Linked by U.S. Gulf Coast refining economics, connecting crude feedstock costs to distillate output through crack spreads
- **WTI-Brent:** Linked by trans-Atlantic crude arbitrage, with price differentials constrained by tanker shipping costs, export capacity, and physical flows.

Table 7 Economic rationale underlying the statistical arbitrage spread universe.

Appendix D: Statistical Arbitrage Cointegration Tests

Spread	Cointegration T-Stat	p-Value	Cointegration Rejection (5%)
Ethane-Natgas	-2.38	0.33	FALSE
Propane-Ethane	-3.85	0.01	TRUE
Propane-Butane	-3.34	0.05	TRUE
RBOB-Butane	-4.52	0.00	TRUE
RBOB-Brent	-4.75	0.00	TRUE
Brent-ULSD	-2.62	0.23	FALSE
ULSD-WTI	-3.02	0.11	FALSE
WTI-Brent	-2.71	0.19	FALSE

Table 8 Engle–Granger cointegration test statistics, p values, and rejection indicators at the 5% significance level.

Appendix E: Unit Conversions to Dollars per Barrel

Commodity	Bloomberg Code	Exchange Quotation	Conversion to $/bbl	Multiplier Used
Propane	BAP	¢/gal	$/bbl = (¢/gal) × 100 × 0.42	0.42 × 100
Ethane	CAP	¢/gal	$/bbl = (¢/gal) × 100 × 0.42	0.42 × 100
Butane	DAE	¢/gal	$/bbl = (¢/gal) × 100 × 0.42	0.42 × 100
RBOB	XB	$/gal	$/bbl = $/gal × 42	0.42
ULSD	HO	$/gal	$/bbl = $/gal × 42	0.42
WTI	CL	$/bbl	Already in $/bbl	1.00
Brent (ICE)	CO	$/bbl	Already in $/bbl	1.00
Gasoil (ICE)	QS	$/metric ton	$/bbl ≈ $/ton ÷ 7.45	1 ÷ 7.45 ≈ 0.134228
Natgas	NG	$/MMBtu	$/bbl-equiv = $/MMBtu × 5.8	5.80

Table 9 All commodities are from CME except Brent & Gasoil

Appendix F: Momentum, Carry, and Value on Individual Commodities Performance

Commodity		Full Sample			2015-2022			2022-2025		
Strategy		MOM	CAR	VAL	MOM	CAR	VAL	MOM	CAR	VAL
WTI	Return	-2%	6%	2%	0%	11%	2%	-6%	-2%	3%
	Vol	15%	23%	17%	13%	24%	16%	19%	20%	17%
	IR	-0.12	0.28	0.13	0.03	0.46	0.10	-0.31	-0.11	0.18
Brent	Return	-9%	5%	2%	0%	7%	2%	-26%	0%	4%
	Vol	28%	16%	17%	23%	16%	17%	36%	15%	18%
	IR	-0.33	0.30	0.14	-0.01	0.46	0.10	-0.72	-0.01	0.21
ULSD	Return	-1%	8%	3%	-1%	7%	3%	-2%	8%	4%
	Vol	50%	20%	21%	53%	19%	19%	44%	22%	25%
	IR	-0.02	0.37	0.15	-0.01	0.39	0.14	-0.05	0.35	0.16
RBOB	Return	2%	9%	4%	-4%	8%	2%	13%	10%	8%
	Vol	33%	76%	24%	36%	91%	24%	26%	32%	25%
	IR	0.06	0.12	0.16	-0.11	0.09	0.06	0.49	0.30	0.34
Gasoil	Return	6%	9%	2%	7%	6%	2%	3%	13%	3%
	Vol	20%	25%	21%	17%	24%	17%	24%	28%	26%
	IR	0.28	0.34	0.10	0.41	0.26	0.09	0.12	0.45	0.10
Natgas	Return	-5%	3%	1%	-2%	4%	1%	-8%	2%	0%
	Vol	17%	10%	7%	12%	8%	4%	23%	13%	9%
	IR	-0.27	0.31	0.13	-0.21	0.44	0.27	-0.35	0.19	0.03
Propane	Return	5%	2%	2%	8%	5%	3%	1%	-3%	1%
	Vol	6%	7%	7%	7%	7%	7%	5%	6%	8%
	IR	0.83	0.31	0.29	1.16	0.75	0.37	0.12	-0.53	0.16
Butane	Return	8%	3%	2%	10%	7%	3%	5%	-3%	0%
	Vol	7%	9%	9%	8%	9%	9%	6%	9%	9%
	IR	1.09	0.35	0.21	1.25	0.74	0.34	0.75	-0.34	0.00
Ethane	Return	1%	1%	0%	1%	2%	1%	0%	0%	0%
	Vol	5%	6%	3%	4%	5%	2%	5%	7%	4%
	IR	0.12	0.16	0.06	0.15	0.29	0.22	0.08	-0.02	-0.11
EW PORT	Return	0%	5%	1%	2%	6%	2%	-2%	3%	0%
	Vol	13%	15%	11%	13%	16%	11%	14%	12%	12%
	IR	0.03	0.34	0.09	0.15	0.40	0.15	-0.17	0.21	0.01

Table 10: Performance statistics for momentum, carry, and value strategies applied to individual commodities. Reported metrics include average returns mean of log returns, realized volatility as standard deviation of log returns, and information ratio as their fraction over the full sample and split subsamples.

Appendix G: Value Performance Across NGLs and Core-6

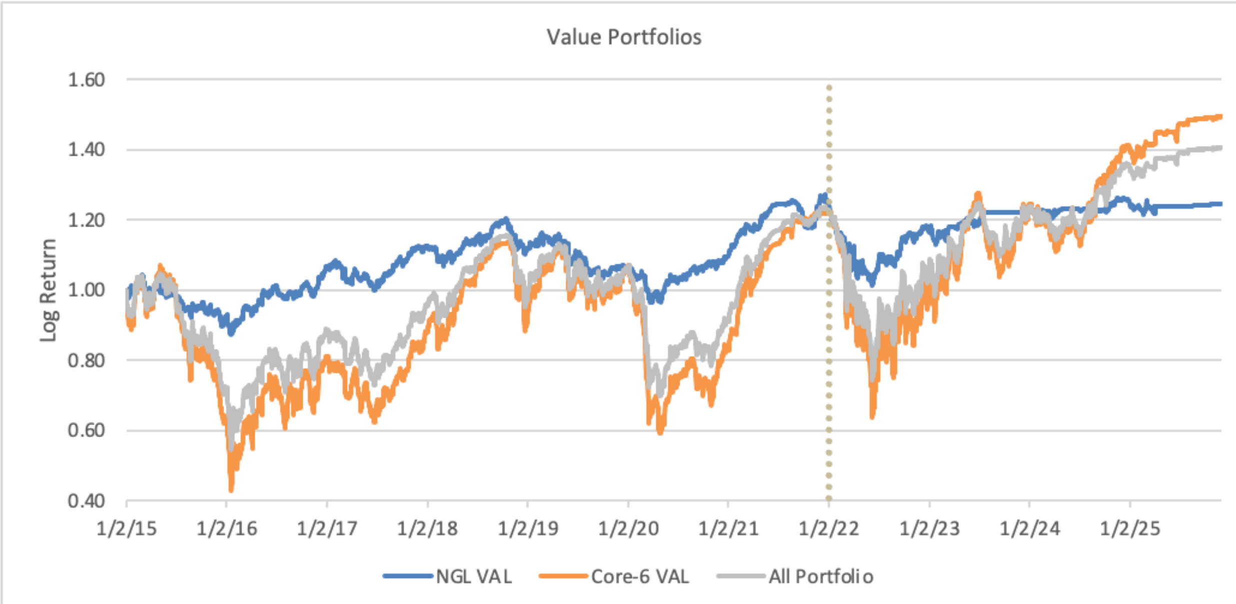

Figure 11: Cumulative log equity curves for value strategies applied to NGL commodities, Core-6 commodities, and the combined portfolio. The vertical dotted line marks the January 2022 sample split.

Appendix H: Statistical Arbitrage Performance Across Individual Spreads

STAT ARB		Full Sample	2015-2022	2022-2025
WTI-Brent	Return	6%	8%	3%
	Vol	9%	11%	4%
	IR	0.65	0.71	0.65
WTI-ULSD	Return	4%	7%	-2%
	Vol	13%	14%	12%
	IR	0.29	0.54	-0.21
Brent-ULSD	Return	0%	2%	-6%
	Vol	10%	9%	12%
	IR	-0.05	0.28	-0.46
RBOB-Brent	Return	6%	8%	3%
	Vol	8%	9%	6%
	IR	0.76	0.88	0.45
RBOB-Butane	Return	14%	17%	8%
	Vol	31%	37%	12%
	IR	0.45	0.46	0.68
Propane-Butane	Return	1%	0%	3%
	Vol	20%	23%	10%
	IR	0.05	0.00	0.31
Propane-Ethane	Return	15%	16%	13%
	Vol	44%	53%	17%
	IR	0.33	0.29	0.78
Ethane-Natgas	Return	10%	8%	14%
	Vol	20%	23%	14%
	IR	0.51	0.37	1.00
EW Port	Return	7%	8%	4%
	Vol	8%	9%	5%
	IR	0.90	0.92	0.95

Table 11 Performance statistics for individual statistical arbitrage spreads. Reported metrics include average return, realized volatility, and information ratio across the full sample and split subsamples, along with the equally weighted spread portfolio.

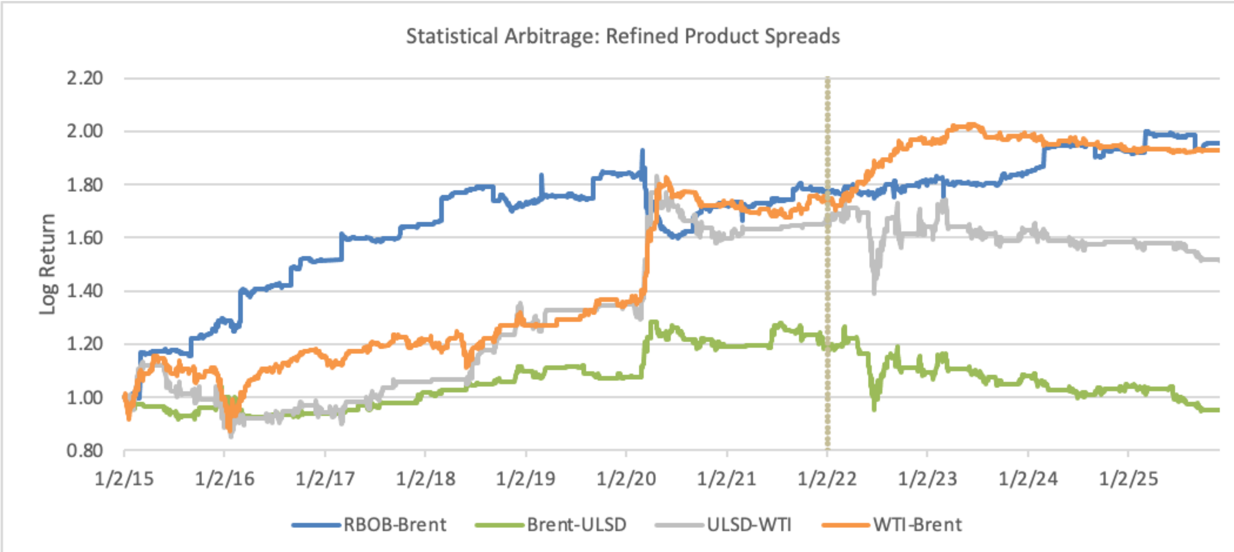

Figure 12 Cumulative log equity curves for refined product statistical arbitrage spreads. The figure illustrates heterogeneous performance across spreads and changing dynamics around the 2022 regime shift.

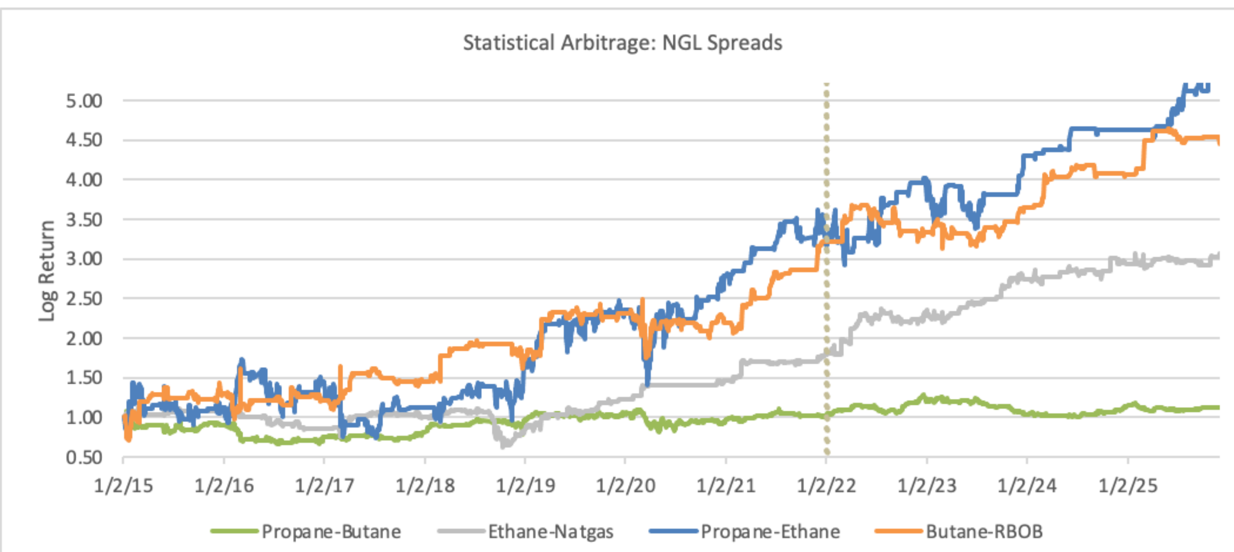

Figure 13 Cumulative log equity curves for natural gas liquid statistical arbitrage spreads. The figure highlights dispersion in spread behavior and the contribution of individual relationships to overall statistical arbitrage performance.

Appendix I: Strategy Correlation Matrices

MOM	WTI	Brent	ULSD	Gasoil	RBOB	Natgas	Propane	Ethane	Butane
WTI		0.77	0.51	0.45	0.41	0.04	0.27	0.05	0.25
Brent	0.77		0.52	0.42	0.44	0.07	0.28	0.06	0.23
ULSD	0.51	0.52		0.35	0.56	0.04	0.29	0.09	0.27
Gasoil	0.45	0.42	0.35		0.31	0.02	0.21	0.06	0.19
RBOB	0.41	0.44	0.56	0.31		0.04	0.21	0.09	0.21
Natgas	0.04	0.07	0.04	0.02	0.04		0.04	0.29	0.04
Propane	0.27	0.28	0.29	0.21	0.21	0.04		0.15	0.70
Ethane	0.05	0.06	0.09	0.06	0.09	0.29	0.15		0.12
Butane	0.25	0.23	0.27	0.19	0.21	0.04	0.70	0.12	

Figure 14 Pairwise correlations of log returns across commodities for the momentum strategy. Correlations are computed using strategy level log returns over the full sample period.

CAR	WTI	Brent	ULSD	Gasoil	RBOB	Natgas	Propane	Ethane	Butane
WTI		0.89	0.72	0.43	0.59	0.07	0.28	0.10	0.25
Brent	0.89		0.75	0.43	0.59	0.06	0.34	0.12	0.30
ULSD	0.72	0.75		0.38	0.72	0.06	0.26	0.12	0.22
Gasoil	0.43	0.43	0.38		0.30	0.04	0.07	0.05	0.05
RBOB	0.59	0.59	0.72	0.30		0.04	0.20	0.08	0.17
Natgas	0.07	0.06	0.06	0.04	0.04		0.08	0.42	0.07
Propane	0.28	0.34	0.26	0.07	0.20	0.08		0.15	0.72
Ethane	0.10	0.12	0.12	0.05	0.08	0.42	0.15		0.14
Butane	0.25	0.30	0.22	0.05	0.17	0.07	0.72	0.14	

Figure 15 Pairwise correlations of log returns across commodities for the carry strategy. Correlations are computed using strategy level log returns over the full sample period.

VAL	WTI	Brent	ULSD	Gasoil	RBOB	Natgas	Propane	Ethane	Butane
WTI		0.96	0.87	0.75	0.65	0.11	0.64	0.14	0.62
Brent	0.96		0.90	0.75	0.68	0.11	0.65	0.16	0.65
ULSD	0.87	0.90		0.71	0.73	0.12	0.60	0.17	0.59
Gasoil	0.75	0.75	0.71		0.53	0.09	0.41	0.10	0.39
RBOB	0.65	0.68	0.73	0.53		0.07	0.47	0.12	0.47
Natgas	0.11	0.11	0.12	0.09	0.07		0.09	0.36	0.08
Propane	0.64	0.65	0.60	0.41	0.47	0.09		0.30	0.87
Ethane	0.14	0.16	0.17	0.10	0.12	0.36	0.30		0.29
Butane	0.62	0.65	0.59	0.39	0.47	0.08	0.87	0.29	

Figure 16 Pairwise correlations of log returns across commodities for the value strategy. Correlations are computed using strategy level log returns over the full sample period.